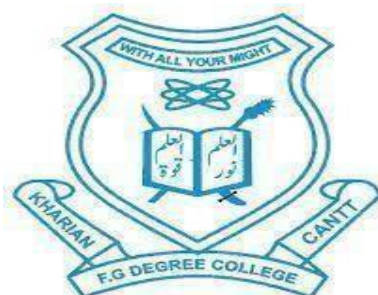




Final Year Design Project

KIPS COACHING KHARIAN



By

Raima Nasir 2022-fkw-16

Maryam Iftikhar 2022-fkw-14

Kiran Shahzadi 2022-fkw-30

Nayab Zahoor 2022-fkw-15

Under the supervision of

Ms. Pakiza Saif

Session: 2022-2026

**DEPARTMENT OF INFORMATION TECHNOLOGY
F.G DEGREE COLLEGE FOR WOMEN, KHARIAN CANTT**

**FACULTY OF COMPUTING &
INFORMATION TECHNOLOGY (FCIT), UNIVERSITY OF
THE PUNJAB, LAHORE**



KIPS COACHING KHARIAN

**A project presented to
University of the Punjab, Lahore**

In partial fulfilment of the requirement for the degree of

Session: 2022-2026

By

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Signature: _____

Raima Nasir [080045]

Signature: _____

Kiran Shahzadi [080052]

Signature: _____

Maryam Iftikhar [080035]

Signature: _____

Nayab Zahoor [080039]



CERTIFICATE OF APPROVAL

It is to certify that the final year design project (FYDP) of BSIT “Kips Coaching App Kharian” was developed by **MARYAM IFTIKHAR (2022-Fkw-14), NAYAB ZAHOOOR (2022-FKW-15), RAIMA NASIR (2022-FKW-16) and KIRAN SHAHZADI (2022-FKW-30)**

Under the supervision of “Ms. PAKIZA SAIF” in my opinion it is fully adequate, in scope and quality for the degree of Bachelors of Science in INFORMATION TECHNOLOGY.

Signature: _____

Supervisor

Miss Pakiza Saif

FG Degree College for Women Kharian Cantt

Dated: _____

Signature: _____

HOD, Department of Information Technology

Mrs. Sidra Khan

FG Degree College for Women Kharian Cantt

Dated: _____



PROOFREADING CERTIFICATE

It is certified that this document does not contain any spelling, punctuation or grammatical mistakes. This document is well organized and also meet the defined objectives.

Student Name	Registration Number	Roll no	Signature
Raima Nasir	(2022-fkw-16)	080045	_____
Maryam Iftikhar	(2022-fkw-14)	080035	_____
Kiran Shahzadi	(2022-fkw-30)	080052	_____
Nayab Zahoor	(2022-fkw-15)	080039	_____



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Their support helped us understand the requirements and build a better solution. Finally, we thank all those who helped us directly or indirectly and help us reach this stage.

Signature: -----

Raima Nasir [080045]

Signature: -----

Kiran Shahzadi [080052]

Signature: -----

Maryam Iftikhar [080035]

Signature: -----

Nayab Zahoor [080039]



Abbreviations

FCIT	Faculty of Computing and Information Technology
JSON	JavaScript Object Notation
UI	User Interface
XML	Extensible Markup Language



Table of Contents

Introduction	13
1.Introduction	14
1.1 Problem Statement	14
1.2 Problem Solution	14
1.3 Objectives of the Proposed System	15
1.4 Scope	15
1.5 System Components	15
1.6 Related Systems	18
1.7 Vision Statement.....	19
1.8 System Limitations and Constraints	19
1.9 Technologies and Tools	19
1.10 Project Deliverables	20
1.11 Project Planning	21
1.12 Summary	21
Literature Review	22
2. Review.....	23
2.1 Literature Survey.....	23
2.2 Related Work	23
2.3 Gap Analysis	24
2.4 Summary	24
Requirements Analysis.....	25
3.Analysis.....	26
3.1 User classes and Characteristics.....	26
3.2 Requirement Identifying Techniques	27
3.3 Functional Requirements.....	56
3.4 Non-Functional Requirements	71
3.5 External Interface Requirements	71
3.6 Summary.....	72



Design and Architecture.....	73
4.System Design.....	74
4.1 Design Consideration.....	74
4.2 Requirement Traceability Matrix.....	74
4.3 Design Models.....	77
4.4 Architectural Design	82
4.5 Data Design	84
4.6 User Interface Design.....	97
4.7 Behavioral Model.....	90
4.8 Design Decisions.....	95
4.9 Summary.....	97
4.10References.....	97



List Of Figures

Figure 1: Gantt Chart	21
Figure 2: Use Case Diagram	28
Figure 3: Story boarding	56
Figure 4: Class Diagram	77
Figure 5: Sequence Diagram (Admin).....	78
Figure 6: Sequence Diagram (Student).....	79
Figure 7: Sequence Diagram (Teacher)	80
Figure 8: Sequence Diagram (Parent).....	81
Figure 9: State Transition Diagram.....	82
Figure 10: Architecture Diagram	83
Figure 11: Component Diagram	84
Figure 12: ERD Schema	85
Figure 13:Screen Image	87
Figure 14:Screen Image	88
Figure 15:Screen Image	88
Figure 16: Screen Image	89
Figure 17:Sequence Diagram (Admin).....	91
Figure 18: Sequence Diagram (Student).....	92
Figure 19: Sequence Diagram (Teacher)	93
Figure 20: Sequence Diagram (Parent).....	94
Figure 21: State Transition Diagram.....	95



List of Tables

Table 1: Related Systems.....	18
Table 2: Technologies and Tools.....	19
Table 3: Related Systems.....	23
Table 4: User Classes and Characteristics	26
Table 5 Use Case 1: Register.....	29
Table 6 Use Case 2: Reset password	30
Table 7 Use Case 3: Login.....	31
Table 8 Use Case 4: Approve parent and student registration.....	32
Table 9 Use Case 5: Add/Delete/update student teacher and parent	33
Table 10 Use Case 6: Manage announcements.....	34
Table 11 Use Case 7: View system reports	35
Table 12 Use Case 8: Record fee payment	36
Table 13 Use Case 9: Classes timetable	37
Table 14 Use Case 10: View enrolled subjects.....	38
Table 15 Use Case 11: Attend online classes	39
Table 16 Use Case 12: Download study material	40
Table 17 Use Case 13: View and attempt online quizzes and assignment	41
Table 18 Use Case 14: View fee history and status	42
Table 19 Use Case 15: Notification for fee payment.....	43
Table 20 Use Case 16: Create link for online classes	44
Table 21 Use Case 17: Upload study material.....	45
Table 22 Use Case 18: Upload and view assignment.....	46
Table 23 Use Case 19: Create online quizzes	47
Table 24 Use Case 20: Manage student performance	48
Table 25 Use Case 21: Communication with parents.....	49
Table 26 Use Case 22: Link parent to child.....	50
Table 27 Use Case 23: View child profile	51
Table 28 Use Case 24: View result	52
Table 29 Use Case 25: Receive announcements	53
Table 30 Use Case 26: Feedback and review	54
Table 31 Use Case 27: Logout.....	55
Table 32: Functional Requirement #1.....	58
Table 33: Functional Requirement #2.....	58
Table 34: Functional Requirement #3.....	59
Table 35: Functional Requirement #4.....	59
Table 36: Functional Requirement #5.....	60
Table 37: Functional Requirement #6.....	60
Table 38: Functional Requirement #7.....	61



Table 39: Functional Requirement #8.....	61
Table 40: Functional Requirement #9.....	62
Table 41: Functional Requirement #10.....	62
Table 42: Functional Requirement #11.....	63
Table 43: Functional Requirement #12.....	63
Table 44: Functional Requirement #13.....	64
Table 45: Functional Requirement #14.....	64
Table 46: Functional Requirement #15.....	65
Table 47: Functional Requirement #16.....	65
Table 48: Functional Requirement #17.....	66
Table 49: Functional Requirement #18.....	66
Table 50: Functional Requirement #19.....	67
Table 51: Functional Requirement #20.....	67
Table 52: Functional Requirement #21.....	68
Table 53: Functional Requirement #22.....	68
Table 54: Functional Requirement #23.....	69
Table 55: Functional Requirement #24.....	69
Table 56: Functional Requirement #25.....	70
Table 57: Functional Requirement #26.....	70
Table 58: Functional Requirement #27.....	70
Table 59: Traceability Matrix	74
Table 60: Data Dictionary	86
Table 61: Screen object (Live classes).....	89
Table 62: Screen object (Notifications and announcements)	90
Table 63: Screen object (Fee payment and confirmation)	90
Table 64: Objects and Attributes	96



Chapter 1

Introduction



1.Introduction

This chapter: Introduction contains a thorough overview, including goals, objectives and defined scope of our project. This chapter also represents the technologies and tools that will be used in implementation of project, along with Vision Statement. Importance of this chapter is that it fully explains the functionality and scope of proposed project. Achieving project objectives requires a complete understanding of the problem statement. Failure to do so will result in the objectives not being achieved.

1.1 Problem Statement

The existing system of **KIPS Coaching Kharian** is based on a manual process, which has led to several operational issues. Students face long queues during registration and fee submission due to the absence of an online payment facility. Additionally, there are no online scheduling updates, making the overall process time-consuming and inconvenient for students.

As a result, students are required to visit the coaching center repeatedly to obtain information, which leads to unnecessary time wastage. Teachers also experience difficulties in managing student records, sharing study materials and maintaining effective communication with parents. Furthermore, there is no centralized online platform that connects students, teachers and parents. In certain situations, such as adverse weather conditions like rain, students are unable to attend physical classes. This often results in missing important lectures and tests, further affecting their academic performance.

1.2 Problem Solution

The **KIPS Coaching Kharian** system is designed to address the challenges faced by students, teachers and administrators in managing academic activities. In the existing system, many tasks such as student registration and fee submission are performed manually, which leads to inefficiency and inconvenience.

This application provides a complete online platform where students can register, attend online classes, download study materials, check class schedules, pay fees and view results from anywhere. Teachers can conduct online classes, share notes and other learning resources and communicate effectively with students and parents. The administration can manage student records, monitor academic performance and handle the admission process online.

Overall, the **KIPS Coaching Kharian** system aims to make the learning process faster, more organized and more convenient for all users. It also supports students in different situations by providing uninterrupted access to educational resources.



1.3 Objectives of Proposed System

- To automate the Online Coaching service of KIPS.
- Live classes will conduct through the Google meet links.
- To provide features like online quizzes, result tracking and notification of class starting.
- To make communication between teachers and parents more effective.
- To ensure that students can access study material anytime and anywhere.
- To provide notifications to keep the students and teacher updated.
- To improve the design and layout for better user experience.
- To provide a better dashboard for monitoring student progress.
- To provide user friendly and accessible app for Android devices.

1.4 Scope

Our “**KIPS COACHING KHARIAN**” project is designed to modernize and enhance the learning mobile platform for students and teachers. Our project includes the following features:

Project Scope

- Develop a mobile-based digital learning platform for Matric and Intermediate students.
- Enable online enrollment and secure access for students, teachers and parents.
- Provide sharing of study materials in digital formats (PDFs, documents, images).
- Display student performance reports on a parent dashboard.
- Include messaging and notification features for effective communication.
- Focus only on digital learning features; no physical or classroom-based setup included.

1.5 System Components

System Components of KIPS COACHING KHARIAN are:

Admin Module:

Login

Admin securely log in to the system

Allow them to manage the whole platform

Approve student and parent registration

Review new register requests

Verify the details and approve or reject the request

Generate a unique ID for Student

Add, update and delete student, teacher and parent

Manually creates teacher accounts or edit information

Remove users from the system as needed



Manage Announcements

Post important updates

Announcements are visible to all users

View System Report

Check reports

Help in monitoring and decision making

Record fee Payment

Update the system when a student pays the fees

Ensure fee status is accurate and visible to the students and parents

Student Module:

Registration and Login Module

Users can create an account (using email, phone number)

Existing users can log in securely

Password encryption and recovery options

View Enroll subjects

Check the subjects they enrolled

Students can enroll more than one subject.

Attend online Classes

Student take online classes as per schedule

Classes schedule or time table will be share with students

Download study material

Students access the study material uploaded by the teacher

Material can be saved for the offline study

View and attempt online quizzes and assignments

Students can view assignments.

Students take quizzes created by teachers.

Scores may be automatically calculated after submission.

View results, fee payment and fee history status

Students can view their academic results.

They also check fee status and payment history anytime.

Receive Notifications for fee payment

The student receives instant notifications for fee payment status and due alerts, including paid amount and balance

Receive Announcements

Students receive alerts for new quizzes, assignments, announcements or fees.



Helps them stay updated

Teacher Module:

Login

Teachers securely log in to manage academic tasks.

Login provides access to teacher-specific tools.

Take Classes and Upload study material

Teachers will take classes as per schedule and upload notes, slides and resources for students.

Files become instantly available to enrolled subjects.

Create quizzes

Teachers create quizzes and set questions.

Students can attempt them online through the system.

Create and share link for online classes

Teacher creates a link for online classes.

After creating the link, teacher share this link with the students.

Manage student performance

Teachers view student progress.

They update marks or give feedback accordingly.

View and upload assignments

Teacher upload assignment for students to complete.

They also review submitted student work and update status.

Announcements

View Notifications

Response and communication with parents

Teachers can message to parents regarding student performance or concern.

Help to build strong communication.

Parent Module:

Registration and login

Parents create account and log in.

Login ensure secure access to child-related information.

Link parent to child account

Parents connect their account with their child's profile.

This give them visibility to academic and fee details.

View results

Parents can see the child's marks, performance reports and progress.



Help them to track academic improvement.

Fee payment

Parents can pay the child's fee through the system.

Payment status update immediately.

Fee history status

Parents check past payment and pending dues.

Provide complete transparency of financial records

1.6 Related systems

There are similar applications related to educational management systems that are functioning effectively in the education industry. These include KIPS Virtual App, The Educator Online System and Roots International Learning Platforms. These institutions have successfully implemented digital platforms and achieved notable success in managing academic and administrative operations. But these systems do not provide fully functional system for live classes and not even any dashboard where student and parents can check/view progress report. KIPS Virtual App just provide video lecture. Due to these limitations, our project aims to create a complete online education system where students can attend live classes, access study material and attend classes from home. Additionally, parents will be able to communicate with teachers and monitor their child's academic performance.

Table 1: Related Systems

App name	Weakness	Proposed Solutions
KIPS Virtual App	The KIPS Virtual System provides limited access for students, as it primarily focuses on student enrollment and recorded video lectures. There is no facility for live online classes which reduces real-time interaction between students and teachers.	An integrated system is proposed that provides online classes, student performance tracking and announcement features to improve communication and academic management.
Roots International Learning Platform	High subscription and maintenance cost.	An affordable and scalable solution is proposed for medium-sized academies like KIPS Coaching Academy, reducing costs while supporting future growth.



1.7 Vision Statement

Students at **KIPS Coaching Kharian** currently face several challenges due to the manual system, including long queues for fee submission, lack of timely updates and the need to physically visit the campus to obtain information. To address these issues, the (App) **KIPS Coaching Kharian** is being developed to provide a digital solution that offers simple registration, online fee payment, instant updates, notifications and easy access to class schedules and other important information from anywhere. This application aims to create a more efficient, accessible and student-friendly learning environment.

1.8 System Limitations and Constraints

Limitations

The main limitation of our project is that **KIPS Coaching Kharian** serves only for the Kharian and surrounding. Our app is specifically designed for students of this branch. Another limitation is that the app currently covers only Punjab region data and does not include information for other KIPS branches across Pakistan. The app needs an active internet connection to work properly.

Constraints

The app does not support offline mode, so user must be connected to the internet to access features. The **KIPS COACHING KHARIAN** is only android app.

1.9 Technologies and Tools

Table 2: Technologies and Tools

Tools, Libraries and Technology	Tools	Version	Rationale
	Android studio	Narwhal	Used for design, code, test and debug
	Libraries	Version	Rationale
	Firebase	3.x	Cloud-based database management system



	Technology	Version	Rationale
	Java	8	Programming language for building: backend and logic of application
	JSON	N/A	Used for structuring all data sent to and received from Firebase
	XML	1.0	For designing the interface of the app
	Node.js	N/A	For server-side logic, Firebase functions and payments gateway setup

1.10 Project Deliverables

Project Deliverables:

Project Proposal: initial document covering basic detail of the project i.e., scope, objectives and planning.

SRS (Software Requirement Specification): This includes all the functional and non-functional requirements that the project will cover.

Design Specification Document: Includes the overall system design and architecture, including User Interfaces and data design as well.

Prototypes: Dummy working project.

Final Project: Fully developed mobile application of our Service.

Admin Panel: A panel/dashboard for the admin to manage and view details.

Testing Reports: Outlining test cases and their results.

Project Report: Complete FYP documentation enlisting all the specifications of our system.

Presentation: A final presentation of our application.



1.11 Project Planning

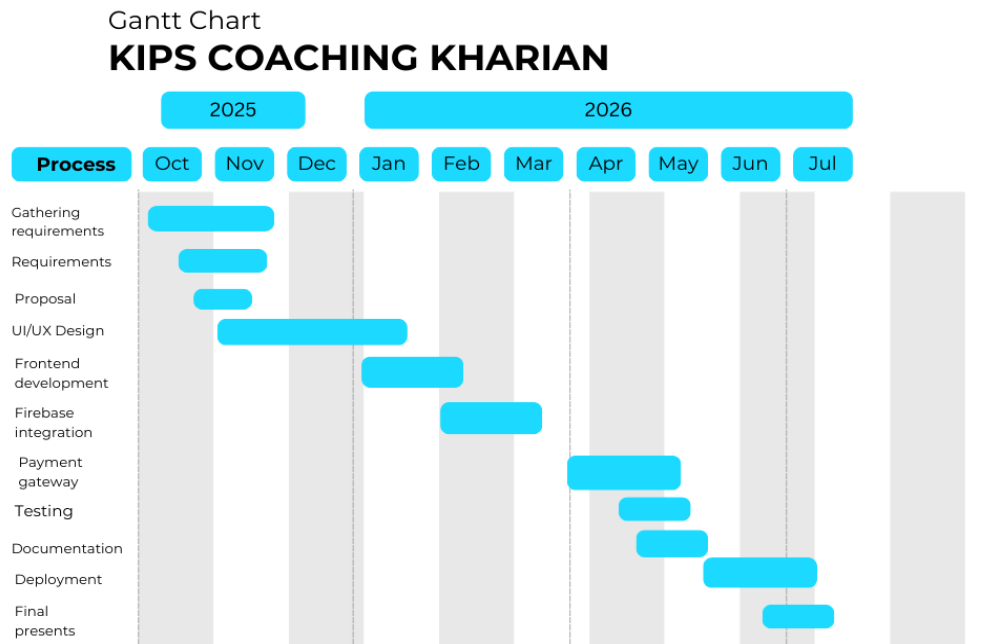


Figure 1: Gantt Chart

1.12 Summary

This chapter introduces the functionalities and core features of our mobile application. First, it states the problem that led to the development of our mobile application followed by the solution which is our app named **KIPS Coaching Kharian**. This chapter also outlines the goals and objectives of our proposed system, along with the risks, limitations and dependencies in details. The scope of our project is included and the system components are clearly defined. A literature review is provided to analyze related systems. We crafted a vision statement that conveys the long-term purpose, intent and significance of our product. Additionally, this chapter list the limitations and constraints of our system and describe the technologies and tools used in the project's implementation. To represent plans for carrying out project activities and showcasing milestones, a Gantt chart is also displayed.



Chapter 2

Literature Review



2.Review

Literature Review of our project highlights the previous researches, systems and technologies related to our system. This review helps by providing the modern recommendations and upgrade the system.

2.1 Literature Survey

Recent studies on Learning management system (LMS) shows that digital platforms improve learning efficiency by offering online content, quizzes, scheduling and progress tracking. Research highlights that mobile based system are more effective in developing countries because they reduce physical barriers such as distance and weather etc. For such system to succeed features like user authentication, content management and fee deposits are essential. However, challenges such as poor internet access, limited teacher training and lack of integrated systems for students, parents and teachers still affect adoption in many educational institutions.

2.2 Related work

As there is a manual system, there is no online system of **KIPS Coaching Kharian**. However, there are similar applications related to educational management systems that are functioning effectively in the education industry. These include KIPS Virtual App, The Educators Online System and Roots International Learning Platforms. These institutions have successfully implemented digital platforms and achieved notable success in managing academic and administrative operations.

Table 3: Related Systems

App name	Weakness	Proposed Solutions
KIPS Virtual App	The KIPS Virtual System provides limited access for students, as it primarily focuses on student enrollment and recorded video lectures. There is no facility for live online classes which reduces real-time interaction between students and teachers.	An integrated system is proposed that provides online classes, student performance tracking and announcement features to improve communication and academic management.



Roots International Learning Platform	High subscription and maintenance cost.	An affordable and scalable solution is proposed for medium-sized academies like KIPS Coaching Academy, reducing costs while supporting future growth.
---------------------------------------	---	---

2.3 Gap analysis

The comparison of existing system shows several gaps that **KIPS COACHING KHARIAN** will address:

Lack of integrated Student, Teacher and Parent communication in current systems.

No user friendly branch specific app managing registrations, schedules, study material and fee payments.

Existing systems does not solve problems like online classes, registration or communication with parents about student performance.

Many platforms lack low cost online payment integration, which your system introduce.

2.4 Summary

The literature and related work indicate a strong need for digital academic management systems that are affordable, mobile friendly and suitable for institution. Current portal provides partial digital services but fail to address issues like communication gaps, accessibility and manual work load. Our system provides a digital platform for students, teacher, parent and administrator that supports efficient learning and management.



Chapter 3

Requirements Analysis



3. Analysis

Software requirement specification for our project SRS, which addresses both the functional and non-functional needs of our mobile application, is presented in this chapter. Along with the hardware, software and communication interfaces that will be utilized to carry out this project. It also outlines the requirements for the user and external interfaces. This chapter acts as a point of reference for the project's progress.

3.1 User Classes and Characteristics

Table 4: User Classes and Characteristics

User Class	User Characteristics
Admin	Secure login View and manage all users Add, Update and remove user's account Generate unique ID for Student Create classes timetable Assign teacher to specific classes View all scheduled sessions and Enrollments Enrolled Student report Send announcements/Notification
Student	Users can create an account Edit personal details View enrolled subjects and upcoming sessions Attend online classes Browse subjects offering by institute View and attempt quizzes and assignments Receive alerts for upcoming sessions and assignments/quizzes deadlines
Teacher	Login and profile verification Join classes and deliver lecture Upload study material Create and share link for classes View enrolled students per subjects



	Create quizzes and assignments Upload assignment/quizzes marks
Parent	Registration and login Communication with teachers directly Access Child's dashboard Online fee payment Receive instant alerts Review and Feedback

3.2 Requirement Identifying Techniques

This part enlists various techniques that are used to identify the requirements of our project **KIPS COACHING KHARIAN**.

Use Case Diagram

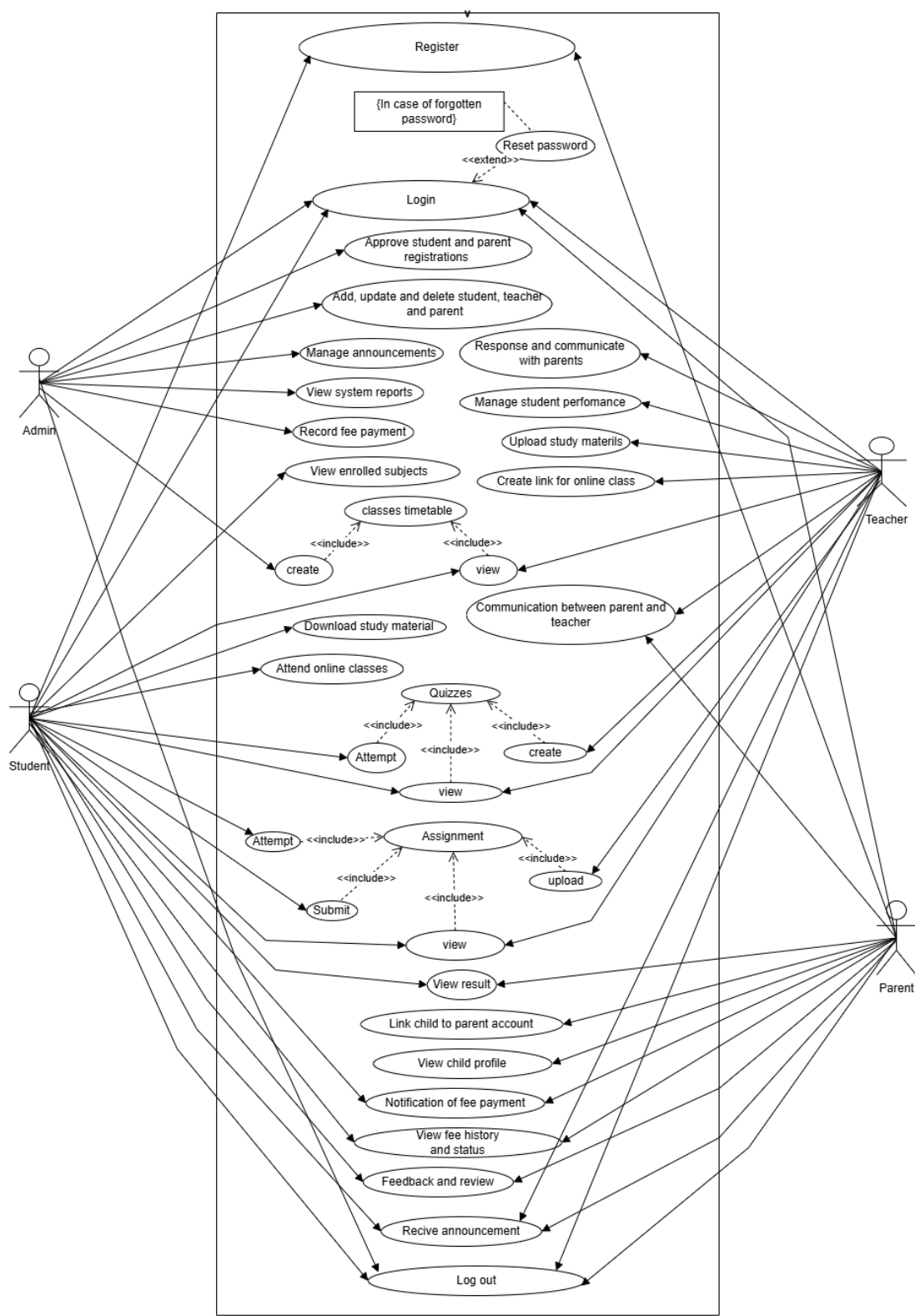




Table 5 Use Case 1: Register

ID:	UC#1
Title	Register
Description	Allow users (student and parent) to register with the KIPS Coaching Kharian app
Primary Actors	Student, Parent
Preconditions	Students and Parents are not already registered and has valid username /email.
Post conditions	User account is created and stored in the system database.
Main success scenario	Users (student and parent) selects “Register”. Users (student and parent) enters required details (Full name, Email, username, Password). After validation, system successfully registers the users (student and parent).
Alternate Flow	Already registered Login
Exception	Invalid email format and weak password
Status	Completed
Priority	Very High
Include	Validate user input



Table 6 Use Case 2: Reset Password

ID:	UC#2
Title	Reset Password
Description	Allows a registered user to reset their forgotten password for the KIPS COACHING KHARIAN APP.
Primary Actors	Student, Admin, Parent
Preconditions	The users (student, admin and parent) must have a registered account within the system
Post conditions	The user's password is successfully reset and they can access their account with the new credentials.
Main success scenario	Users select forgotten password or reset password. User enters their registered email or mobile number. After validation the system sends a password reset link or temporary password. User follow the instructions to set a new password. System successfully updates a user password.
Alternate flow	User enters an unregistered email. Password reset link expires User requests multiple password resets.
Exception	System failure during password update. Emails delivery failure Invalid password format.
Frequency	Whenever a user forgets the password.
Status	Proposed
Priority	High
Include	Validate user input



Table 7 Use Case 3: Login

ID:	UC#3
Title	Login
Description	Allow an authorized user(admin, students, parents and teachers) to securely log in to the system to access their respective functionalities.
Primary Actors	Admin, Parent, Teacher, Student
Preconditions	The user has a valid username and password. The system is operational and the login page is displayed.
Post conditions	The user is successfully authenticated and redirected to the admin dashboard. An error message is displayed for invalid credentials and the user remains on the login page.
Main success scenario	User open the app User enter username/email and password System verify credentials User is logged in successfully
Alternate Flow	User enter incorrect credentials System display an error message
Exception	Server error or network failure during login
Frequency	Weekly or as needed
Status	Draft
Priority	Medium
Include	User authentication Role based access control



Table 8 Use Case 4: Approve student and parent registration

ID:	UC#4
Title	Approve student and parent registrations
Description	Allows an administrator to review and approve registration requests from students and parents
Primary Actors	Admin
Preconditions	Student or parent has submitted a registration
Post conditions	The users (student and parent) registration is approved and their account is activated. Generate a unique ID of Student. A notification is sent to the approved user. The request is remove from the pending list
Main success scenario	Admin log in. Admin view pending registration. Admin select a request. Admin approve the registration. System activate the account and generate a unique key for Student
Alternate Flow	Admin reject the registration request. System send rejection notification.
Exception	System error or connection failure during approval process.
Frequency	As needed
Status	Draft
Priority	Medium
Include	User authentication Notification Service



Table 9 Use Case 5: Add, update and delete students, teachers and parents

ID:	UC#5
Title	Add, update and delete students, teachers and parents
Description	Allow the admin to add new users, update existing users details or delete student, teacher and parent accounts.
Primary Actors	Admin
Preconditions	Admin is logged in and has management privileges.
Post conditions	User records are added, updated or removed from the system database.
Main success scenario	Admin log in Admin select the users(Students, teachers and Parents) . Admin choose add, update or delete option. Admin enter or modify user details. System save changes successfully.
Alternate flow	Admin cancel the operation. No changes are saved.
Exception	Database error or system failure during operation.
Frequency	Weekly
Status	Approved
Priority	High
Include	User Authentication Data Validation



Table 10 Use Case 6: Manage Announcements

ID:	UC#6
Title	Manage announcement
Description	Admin can create, edit and delete system announcement for students, teachers and parents.
Primary Actors	Admin
Preconditions	Admin must be logged in and have permissions to manage announcements.
Post conditions	Announcements are updated and relevant users can view them via notifications.
Main success scenario	Admin log in. Admin open Announcement module. Admin create or edit an announcement. Admin publish the announcement.
Alternate flow	Admin deletes an announcement. System removes it from user view.
Exception	System or database error during announcement management.
Frequency	Weekly or monthly
Status	Approved
Priority	High
Include	User Authentication Notification Service

*Table 11 Use Case 7: View system reports*

ID:	UC#7
Title	View system report
Description	Admin can view various system report in the system.
Primary Actors	Admin
Preconditions	Admin must be logged in.
Post conditions	The system displays the requested report.
Main success scenario	Admin log in. Admin select System Report. Admin choose the report. System generate and display the report.
Alternate flow	Admin applies filters. System update the report view.
Exception	Report generation failure or database error.
Frequency	Weekly
Status	Approved
Priority	Medium
Include	User Authentication Reporting Module



Table 12 Use Case 8: Record Fee Payment

ID:	UC#8
Title	Record fee payment
Description	Admin can save fee payment record from a student in the system.
Primary Actors	Admin
Preconditions	Admin must be logged in and the student record must exist.
Post conditions	The fee payment is recorded in the system and fee paid notification send to student and parents.
Main success scenario	Admin log in. Admin select. Admin search student record. Admin enter payment detail. System save fee record.
Alternate flow	Admin update an existing payment record. System refresh fee status.
Exception	Payment record save failure or database error.
Frequency	Multiple times daily
Status	Approved
Priority	High
Include	User Authentication Fee Management Module

*Table 13 Use Case9: Classes timetable*

ID:	UC#09
Title	Classes timetable
Description	Allow an admin to create and modify weekly class schedule within the system and allow the student and teacher to view the classes timetable.
Primary Actors	Admin
Preconditions	The admin must be logged into the application and the admin have appropriate permission to edit scheduled and teacher and student have permission to view timetable.
Post conditions	The new timetable is saved and published to the system visible to the relevant user student and teacher name.
Main success scenario	User log in to the system. User open timetable section. System retrieves timetable data. System display class timetable.
Alternate flow	Admin change day or view weekly timetable.
Exception	Timetable not available or system error
Frequency	Daily
Status	Approved
Priority	Medium
Include	Login



Table 14 Use Case10: View enrolled subjects

ID:	UC#10
Title	View enrolled subjects
Description	Students view enrolled subjects
Primary Actors	Student
Preconditions	The student should be logged in.
Post conditions	Enrolled subjects displayed on screen.
Main success scenario	Student log in. Student select the subjects. System retrieve enrolled subjects. Subject list is displayed.
Alternate flow	No subject enrolled. System display a message.
Exception	Data retrieval failure or system error
Frequency	Daily
Status	Approved
Priority	Medium
Include	User Authentication Subject Management Module



Table 15 Use Case11: Attend online classes

ID:	UC#11
Title	Attend live classes
Description	Student joins and attend the scheduled online class using the provided link.
Primary Actors	Student
Preconditions	The student is registered and must be logged into the application.
Post conditions	Student successfully attends the online class.
Main success scenario	Student log in. Student view online class schedule. Student clicks on the class meeting link. System redirects to the online meeting platform. Student joins the meeting and attend the class
Alternate flow	Student join the class directly from the notification link. Student joins the class late.
Exception	Invalid or expire link shows an error message. Internet or system issue unable to join the class.
Frequency	Daily or as scheduled
Status	Approved
Priority	High
Include	Authenticate student view class schedule



Table 16 Use Case12: Download Study Material

ID:	UC#12
Title	Download study material
Description	Registered students can download available study materials from the app.
Primary Actors	Student
Preconditions	The student must be logged in and study material must be available.
Post conditions	The selected study material is downloaded to the student's device.
Main success scenario	Student log in. Student select subject. Student choose study material. System downloads the file.
Alternate flow	Student view material online instead of downloading. System displays the content.
Exception	Download failure or network issue.
Frequency	Whenever a student needs to access material offline.
Status	Proposed
Priority	High
Include	User Authentication File Download



Table 17 Use Case13: View and attempt online quizzes and assignment

ID:	UC#13
Title	View and attempt quizzes and assignment
Description	Allow a student to view an available quiz and assignments within the app.
Primary Actors	Student
Preconditions	The student must be registered and logged into the application.
Post conditions	The student must be logged into the system and a quiz and assignment must be available and assigned to them.
Main success scenario	Student log in. Student select Quizzes and assignment. Student select an available quiz and assignment. Student attempt and submits quiz and assignment
Alternate flow	Student exits quiz and assignment before submission. System auto save or discard attempt.
Exception	Timer expiration or network failure during quiz and assignment. .
Frequency	Whenever a student want.
Status	Proposed
Priority	High
Include	User Authentication Quiz and assignment



Table 18 Use Case 14: View fee history and status

ID:	UC#14
Title	View fee history and status
Description	Allow parent to view the fee history and payment status of a linked child's account.
Primary Actors	Parent, student
Preconditions	The parent and the child must have an active account, the parent account must be linked with the child's account and the fee data must exist for the child.
Post conditions	The child's fee history and status are successfully displayed on the parent's dashboard.
Main success scenario	Parent/Student logs into the system. Selects "View Fee History and Status" option. System displays detailed payment history, including dates, amounts and status (paid/unpaid). Parent/Student reviews current balance and pending payments. Optionally, Parent/Student downloads or prints the fee report.
Alternate flow	Parent/Student filters fee history by month or year. Parent/Student receives a summary of total fees paid and pending.
Exception	No fee records found for the student. System error prevents displaying the fee history.
Frequency	Varies (e.g. once per month/year).
Status	Approved
Priority	High
Include	Login, Link parent account to child, notification for fee payment



Table 19 Use Case 15: Notification for Fee payment

ID:	UC#15
Title	Notification for Fee payment
Description	Allow parent to pay their child's fees through the system.
Primary Actors	Parent, student
Preconditions	Automatically notifies parents or students about pending or upcoming fee payments, including due dates and payment status.
Post conditions	Parent or student receives a notification about pending or due fees.
Main success scenario	System checks pending or upcoming fee payments. System generates a notification. System sends notification to parent/student via app, email, or SMS. Parent/student receives the notification. Parent/student views payment details and status.
Alternate flow	Parent/student has multiple children; system sends consolidated notifications. Parent/student views history of previous fee notifications.
Exception	Fee record not found for the student. Notification delivery fails due to network or system error.
Frequency	Monthly.
Status	Approved
Priority	High
Include	Login, Link parent account to child, view fee history



Table 20 Use Case16: Create link for online classes

ID:	UC#16
Title	Create link for online classes
Description	Teacher creates and shares an online class link with the students.
Primary Actors	Teacher
Preconditions	The teacher must be logged into the application.
Post conditions	Online classes link is created and available to the students.
Main success scenario	Teacher log in. Teacher select create online Class. Teacher enter class detail (subject, date, time). Teacher enters meeting link. Teacher click saves/ share. System saves the link and notifies students.
Alternate flow	Teacher edit class detail. Teacher reschedule the class.
Exception	Invalid or missing meeting link error shown. System or internet failure.
Frequency	Daily or as needed
Status	Approved
Priority	High
Include	Authenticate teacher notify students



Table 21 Use Case17: Upload study material

ID:	UC#17
Title	Upload study material
Description	Allow a teacher to upload study material for students within the app.
Primary Actors	Teacher
Preconditions	The teacher must be logged into the application. The teacher must have a course assigned to them.
Post conditions	Study material must be available for students to view.
Main success scenario	Teachers log in. Teacher select Study Materials. Teacher upload file and add details. System validate and save the material.
Alternate flow	Teacher update or delete existing material. System reflect changes.
Exception	File upload failure or unsupported file format
Frequency	Weekly or as needed
Status	Approved
Priority	High
Include	User Authentication File Management



Table 22 Use Case18: Upload and view assignments

ID:	UC#18
Title	Upload and view assignments
Description	Teachers upload assignment and view assignment for their classes.
Primary Actors	Teacher
Preconditions	The teacher must be logged in.
Post conditions	Assignment details displayed on screen or confirmation of upload received.
Main success scenario	Teacher select subject. Teacher click upload assignment. Teacher open assignment section. System display uploaded assignments
Alternate flow	Teacher edit, update or reload the assignment.
Exception	Invalid file format, upload failure or assignment not found
Frequency	Weekly or as needed
Status	Approved
Priority	Medium
Include	Log in



Table 23 Use Case19: Create online quizzes

ID:	UC#19
Title	Create quizzes
Description	Allow a teacher to create quizzes within the system.
Primary Actors	Teacher
Preconditions	The teacher must be logged into the application. The teacher must have a course assigned to them.
Post conditions	Study material must be available for students to view.
Main success scenario	Teacher log in. Teacher select Quizzes. Teacher create quiz questions and settings. Teacher will set the quiz timer. Teacher publish the quiz.
Alternate flow	Teacher edit or delete a quiz. System update quiz details.
Exception	Quiz save failure or system error.
Frequency	Weekly or as needed
Status	Approved
Priority	High
Include	User Authentication Quiz Management

*Table 24 Use Case 20: Manage student performance*

ID:	UC#20
Title	Manage student performance
Description	Allow teacher to record, update and view student grades and performance metrics.
Primary Actors	Teacher
Preconditions	The teacher must be logged in and assigned to relevant courses.
Post conditions	Student performance data is recorded and updated in the system.
Main success scenario	Teacher log in to the system. Teacher select class or subject. Teacher view student performance list. Teacher enter or update marks/grades. System save and display updated performance.
Alternate flow	Teacher edit or update previously entered performance data.
Exception	Invalid data entry. Student record not found or system error.
Frequency	Weekly or as needed
Status	Approved
Priority	High
Include	Login



Table 25 Use Case21: Communication with Parents

ID:	UC#21
Title	Communication with Parents
Description	System allow teachers to send messages or update to parents and parents to view and respond to these communication.
Primary Actors	Teacher, Parent
Preconditions	The teacher or parents must be logged in.
Post conditions	Message or notification is delivered to parents.
Main success scenario	Teacher logs in to the system. Teacher selects student or parent Teacher write a message or announcements. Teacher sends a message. System delivers the message to the parent through the email. Parent views the message.
Alternate flow	Teacher edits or resends the message before final submission. Parent views the message later if not online.
Exception	Parent account not found. Message sending fails due to system error.
Frequency	As needed
Status	Approved
Priority	High
Include	Login



Table 26 Use Case22: Link parent account to child

ID:	UC#22
Title	Link child to parent
Description	Allow child to link their account to a parent existing account.
Primary Actors	Parent
Preconditions	The parent and the child must have an active account.
Post conditions	The child account is successfully linked to the parent account.
Main success scenario	Parent logs in to the system. Parent selects “Link child account” option. Parent enters child unique ID or registration number. System verifies the child account. System links the child account to the parent account. Child receives confirmation of successful linking.
Alternate flow	Parent requests linking for multiple children at once. Parent updates or removes an existing linked child account.
Exception	Child account not found. Parent provide incorrect child ID. System error prevents linking.
Frequency	As needed
Status	Approved
Priority	High
Include	Login



Table 27 Use Case23: View child profile

ID:	UC#23
Title	View child profile
Description	Allow parent to view the profile information of a linked child's account.
Primary Actors	Parent
Preconditions	The parent and the child must have an active account and the parent account must be linked with the child's account.
Post conditions	The child's profile detail is successfully displayed to the parent.
Main success scenario	Parent logs into the system. Parent selects "View Child Profile" option. System displays the child's personal details. System displays academic information, attendance records, and performance metrics. Parent reviews the information.
Alternate flow	Parent navigates between multiple children if more than one child is linked. Parent updates notification preferences while viewing profile.
Exception	No child account linked to parent. System error prevents loading the profile.
Frequency	Frequent/Daily
Status	Approved
Priority	High
Include	Login, Link parent account to child



Table 28 Use Case 24: View result

ID:	UC#24
Title	View result
Description	Allow parent and students to view the academic result.
Primary Actors	Parent, Students
Preconditions	The parent and the child must have an active account. The parent account must be linked with the child's account and the child must have result available in the system.
Post conditions	The child's academic results are successfully displayed to the parent and student dashboard.
Main success scenario	Parent/Student logs into the system. Selects "View Results" option. System displays the student's exam results and grades. System shows performance trends, total marks and subject-wise marks. Parent/Student reviews and optionally downloads the results.
Alternate flow	Parent/Student selects a specific class to view results. System provides graphical representation of performance.
Exception	Results not yet available. Parent/Student account not linked correctly. System error prevents loading results.
Frequency	Frequent/After each exam
Status	Approved
Priority	High
Include	Login, Link child account to parent



Table 29 Use Case25: Receive announcements

ID:	UC#25
Title	Receive announcement
Description	Allows users (Student, Teachers, Parents) to view available announcements within the app through notification.
Primary Actors	Student, Teachers, Parents
Preconditions	The user (Student, Teachers, Parents) must be logged into the application.
Post conditions	Announcement must be available and assigned to them.
Main success scenario	User logs in to the system. User open the announcement section. System retrieves announcements. System display announcements to the user.
Alternate flow	User refresh or filter announcement
Exception	No announcement or system error
Frequency	As needed
Status	Approved
Priority	Medium
Include	Login

*Table30 Use Case 26: Feedback and Review*

Id	UC#26
Title	Feedback and Review
Description	Allows student and parent to give feedback and review about the system
Actor	Parent, student
Preconditions	Parent and student must enter feedback
Post conditions	Parent and student can give feedback and reviews of using app.
Main success scenario	Student and parent give review about system(App).
Alternate Flow	None
Exceptions	Duplicate Review Submission Failure
Status	Completed
Priority	Medium
Include	None



Table 31 Use Case 27: Logout

ID:	UC#27
Title	Logout
Description	The user ends their session and logout of the system.
Primary Actors	Admin, Students, Teachers, Parents
Preconditions	The user is currently logged in.
Post conditions	The user is logged out and the system displays the login screen.
Main success scenario	User clicks the “Logout” button. System terminates the user session. System redirects the user to the login page. User confirms they have successfully logged out.
Alternate flow	User session times out automatically due to inactivity, triggering automatic logout. User cancels logout confirmation (if system asks for confirmation).
Exception	System error prevents logout (rare). Network interruption during logout may require user to retry.
Frequency	Ever session
Status	Approved
Priority	High
Include	Login



Storyboarding

This story board is made using Canva. It depicts our student module in visual form.

KIPS COACHING KHARIAN



Figure 3: Story boarding

3.3 Functional Requirements

Requirements that must be fulfilled to complete the functionality of our system.

- 1.User Registration:** User (student and parent) shall be able to register themselves by providing credentials.
- 2.Reset Password:** User can reset password in the case of forgot password.
- 3.User Login:** User shall be able to login to the system after registration.



- 4. Approve student and parent registration:** Admin shall approve the student and parent registration and manually add the teacher.
- 5. Add, update and delete users:** Admin shall add, update and delete users (student, teacher and parents).
- 6. Manage announcements:** Admin shall manage the announcements for the students, teachers and parents.
- 7. View System Report:** Admin shall view the system report.
- 8. Record fee payment:** Admin shall record fee payment in the system.
- 9. View enrolled subjects:** Student shall view the enrolled subjects.
- 10. Attend live classes:** Students shall attend live classes
- 11. Download study material:** Students shall download the study material.
- 12. View and attempt quizzes and assignment:** Students shall view and attempt quizzes and assignments.
- 13. Upload Study material:** Teacher shall upload the study material.
- 14. Create quizzes:** Teacher shall create quizzes for the students.
- 15. Upload and view assignments:** Teacher shall upload and view assignments.
- 16. Create link for online classes:** Teacher shall create a link for online class.
- 17. Receive announcement:** Users (students, teachers and parents) shall receive announcements.
- 18. Classes timetable:** Admin shall provide the timetable and the teachers and students shall view the classes timetable.
- 19. Manage student performance:** Teacher shall manage the student performance.
- 20. Communication with parents:** Teacher shall communicate with the parents through the email.
- 21. Link parent account to child:** The parent account shall link with the child.
- 22. View child profile:** Parents shall view the profile information of the linked child's account.
- 23. View result:** The students and parents shall view the results.
- 24. Notification for fee payment:** Allow parents and students to pay their fee.
- 25. View fee history and status:** The parent shall view the fee history and payment status of linked child's account.
- 26. Feedback and Review:** Students and Parents give the review about the app performance.
- 27. Logout:** The user ends their session and logout of the system.



3.3.1 Functional Requirements

Table 32 Functional Requirement#1

Identifier	Functional Requirement#1
Title	Register
Requirement	User can register for an account using a unique ID and personal details.
Source	KIPS management
Rationale	To allow users to access the system and its features.
Business Rule (if required)	User must have a valid ID.
Dependencies	None
Priority	High

Table 33 Functional Requirement#2

Identifier	Functional Requirement#2
Title	Reset password
Requirement	System allows users to securely reset their forgotten password.
Source	System requirement
Rationale	To allow users to regain access to the system if they forget their credentials.
Business Rule (if required)	User must provide a valid email address or username to receive a reset link or code. The reset token should expire within a short period and be single use.
Dependencies	User authentication system, email service.
Priority	High



Table 34 Functional Requirement#3

Identifier	Functional Requirement#3
Title	Log in
Requirement	System allows users to securely login into the system
Source	System requirement
Rationale	To provide secure access for system users.
Business Rule (if required)	User must be an admin, student, teacher or parent and login with valid credentials
Dependencies	User login system user authentication
Priority	High

Table 35 Functional Requirement#4

Identifier	Functional Requirement#4
Title	Approve student and parent registration
Requirement	System must verify the student and parent identity during the registration process to ensure authenticity and prevent fraud
Source	System design specification legal compliance
Rationale	To maintain data integrity security and comply with regulations
Business Rule (if required)	User must provide a valid e-mail address and a unique identification document
Dependencies	E-mail system, ID verification services or database, user database
Priority	High



Table 36 Functional Requirement#5

Identifier	Functional Requirement#5
Title	Add, update and delete users
Requirement	Allows an admin to securely add a new users account to the system
Source	kips requirements and parent feedback
Rationale	To efficiently enroll new users and manage records
Business Rule (if required)	User must be logged in and have appropriate permissions and requirement fields must be completed
Dependencies	User login, user database, account creation system
Priority	High

Table 37 Functional Requirement#6

Identifier	Functional Requirement#6
Title	Manage announcement
Requirement	The system shall allow administrators to create, update and delete announcements.
Source	System user
Rationale	To communicate important information to users effectively and maintain current announcements.
Business Rule (if required)	Only administrators can manage announcements.
Dependencies	Database connection
Priority	High



Table 38 Functional Requirement#7

Identifier	Functional Requirement#7
Title	View system reports
Requirement	The system shall allow administrators to view system reports.
Source	System user
Rationale	To monitor system performance and usage.
Business Rule (if required)	Only administrator can view the report.
Dependencies	Database connection, reporting modules.
Priority	High

Table 39 Functional Requirement#8

Identifier	Functional Requirement#8
Title	Record fee payment
Requirement	The system shall allow authorized users to record and manage student fee payments.
Source	Only the Administrator can record the payment.
Rationale	To maintain Accurate and current financial records.
Business Rule (if required)	Only the administrator can record a payment.
Dependencies	Database connection
Priority	High



Table 40 Functional Requirement#9

Identifier	Functional Requirement#9
Title	View Enrolled Subjects
Requirement	Students can view a list of all subjects they are currently enrolled in.
Source	KIPS management
Rationale	To allow students to track their academic progress and access subject-specific content.
Business Rule (if required)	Student must be actively enrolled in at least one subject.
Dependencies	FR#2(Student Login)
Priority	High

Table 41 Functional Requirement#10

Identifier	Functional Requirement#10
Title	Attend online classes
Requirement	The system shall allow students to view the scheduled online and join the class using the provided meeting link at the scheduled time.
Source	KIPS requirement, student feedback
Rationale	Student need a reliable way to attend online classes.
Business Rule (if required)	Student must be registered in the class. Meeting link is accessible only during class time. Stable internet connection is required.
Dependencies	FR#1
Priority	High



Table 42 Functional Requirement#11

Identifier	Functional Requirement#11
Title	View and attempt Quizzes and assignments
Requirement	Students accesses and attempts interactive quizzes and assignments within the system.
Source	KIPS requirement, App Features
Rationale	To allow students to test their understanding and self-access effectively.
Business Rule (if required)	Student must be registered and login.
Dependencies	FR#9(view enrolled subjects)
Priority	High

Table 43 Functional Requirement#12

Identifier	Functional Requirement#12
Title	Download Study Material
Requirement	Students logs in and downloads study material in PDF format.
Source	KIPS requirement, Student Feedback
Rationale	To provide students with offline access to educational content.
Business Rule (if required)	Student must be registered and login.
Dependencies	FR#5(View Quizzes/Tests)
Priority	High



Table 44 Functional Requirement#13

Identifier	Functional Requirement#13
Title	Announcements
Requirement	System sends announcements to students about classes, quizzes and result.
Source	KIPS requirement, Student Feedback
Rationale	To keep students informed and engaged with their academic progress.
Business Rule (if required)	Student must be enrolled in a course.
Dependencies	FR#7(View Fee Status)
Priority	High

Table 45 Functional Requirement#14

Identifier	Functional Requirement#14
Title	Classes Timetable
Requirement	Display a weekly timetable for each class. Showing the subjects, time, date and assigned teacher.
Source	Kips requirement
Rationale	Students and teachers need a clear schedule to plan their activities.
Business Rule (if required)	Classes can't be scheduled during lunch breaks.
Dependencies	FR#1 (Teacher Login)
Priority	High



Table 46 Functional Requirement#15

Identifier	Functional Requirement#15
Title	Upload Study Material
Requirement	Upload study material in pdf form to a specific course or class.
Source	KIPS requirement, teachers feedback
Rationale	Teachers need a mechanism to share resources with students efficiently.
Business Rule (if required)	Only teachers can access the upload functionality.
Dependencies	FR#2 (Class Timetable)
Priority	High

Table 47 Functional Requirement#16

Identifier	Functional Requirement#16
Title	Create Quizzes
Requirement	Create new quizzes, add questions, set time limits and upload them to specific class
Source	KIPS requirement
Rationale	Teachers need a tool to assess student knowledge efficiently.
Business Rule (if required)	At least 2 questions must and maximum time 30 minutes and maximum file size is 50MB.
Dependencies	FR#3 (Upload Study Material)
Priority	High



Table 48 Functional Requirement#17

Identifier	Functional Requirement#17
Title	Create link for online class
Requirement	System shall allow the teacher to create an online class by entering class detail and adding a link and share it with students.
Source	KIPS requirement
Rationale	Teacher need an easy way to schedule and conduct online classes.
Business Rule (if required)	Meeting link must be valid. Class date and time cannot be in the past. Only assigned can create links for their classes.
Dependencies	FR#1
Priority	High

Table 49 Functional Requirement#18

Identifier	Functional Requirement#18
Title	Manage Student Performance
Requirement	Record student performance data (grades, comments) for specific assignment.
Source	KIPS requirement, teacher feedback.
Rationale	To track student progress and provide feedback.
Business Rule (if required)	Data entry must be validated against defined grade scales.
Dependencies	FR#4 (Create Online Quizzes)
Priority	High



Table 50 Functional Requirement#19

Identifier	Functional Requirement#19
Title	Communication with Parents
Requirement	Send text messages or emails to parents of students.
Source	KIPS requirement, parent feedback
Rationale	Timely communication between teachers and parents regarding progress, behavior and announcements.
Business Rule (if required)	Messages must be logged.
Dependencies	FR#5 (Manage Student Performance)
Priority	High

Table 51 Functional Requirement#20

Identifier	Functional Requirement#20
Title	Upload and view Assignments
Requirement	Upload an assignment to students and view previous assignments.
Source	KIPS requirement, teacher feedback
Rationale	Mechanism to distribute assignments to students and manage them within the system.
Business Rule (if required)	Maximum file size is 50MB and only PDF file format.
Dependencies	FR#6 (Communication with Parents)
Priority	High



Table 52 Functional Requirement#21

Identifier	Functional Requirement#21
Title	Link parent to child
Requirement	System allows parents to securely link their account to a child's account.
Source	KIPS requirement, parent Feedback
Rationale	To ensure proper account management and parental oversight.
Business Rule (if required)	Parent and child must both have valid accounts
Dependencies	Parent and child account registration requirements
Priority	High

Table 53 Functional Requirement#22

Identifier	Functional Requirement#22
Title	View child profile
Requirement	System allows parents to securely view their child's profile information
Source	KIPS requirement, parent Feedback, system design
Rationale	Allow parents to monitor and manage their child's information and settings within the system
Business Rule (if required)	Parent must be logged in and linked to the child's profile.
Dependencies	Parent login, child profile creation, link accounts
Priority	High



Table 54 Functional Requirement#23

Identifier	Functional Requirement#23
Title	View results
Requirement	System allows parents and students to securely view their academic results
Source	KIPS requirement, parent Feedback
Rationale	To provide parents with timely updates on their child's academic progress.
Business Rule (if required)	Parent must be logged in and have a child associated with their account.
Dependencies	Payroll login requirement, student result data availability
Priority	High

Table 55 Functional Requirement#24

Identifier	Functional Requirement#24
Title	Notification for Fee payment
Requirement	System allows parent to securely pay fees for their children
Source	KIPS requirement, parent Feedback
Rationale	To provide a convenient and secure method for parents to settle financial obligations and ensure timely payments
Business Rule (if required)	Parent must have a valid username and password and a valid payment method
Dependencies	Parent login requirement, payment gateway integration
Priority	High



Table 56 Functional Requirement#25

Identifier	Functional Requirement#25
Title	View fee history and status
Requirement	System allows parents to securely view their child's payment history and current status.
Source	KIPS requirement, parent Feedback, Student feedback
Rationale	To ensure transparency in financial matters and allow parents to track payments
Business Rule (if required)	Parent must be logged in and linked to a student account
Dependencies	Parent login, student account, fee payment system
Priority	High

Table 57 Functional Requirement #26

Identifier	Functional Requirement #26
Title	Give Feedback and Review
Requirement	Student and parent shall provide feedback and reviews about their experience.
Source	Student, parent
Rationale	Improve System quality
Business Rule	Inappropriate review must be removed
Dependencies	None
Priorities	Medium

Table 58 Functional Requirement#27

Identifier	Functional Requirement#27
Title	Log out
Requirement	System allows users to securely log out from this session
Source	KIPS requirement, Users Feedback



Rationale	To ensure account security and privacy and manage system access
Business Rule (if required)	None
Dependencies	User login requirements
Priority	High

3.4. Non-functional requirements

The non-functional requirements of **KIPS COACHING KHARIAN** are followed below:

3.4.1. Reliability requirements

The system must ensure minimum downtime and available to the users during operational hours. Users need easy access to the platform for all coaching activities. A reliable system ensures continuity of service during all scheduled operational hours.

3.4.2. Usability requirements

The interface must be simple, easy to navigate and used on mobile devices. Clear error message will guide users that is human readable message (e.g. username you entered is not recognized. Please check your spelling).

3.4.3. Performance requirements

The system will load quickly and response in 4 or 5 second if the user has good internet connection. It supports the users to perform their tasks.

3.4.4. Security requirements

All the user's data including personal information, academic records and personal information must be stored securely. The system should have clear access controls defining which action each type of user can perform.

3.5. External interface requirements

The external interface requirements of our system are given below:

3.5.1. User interface requirements



The application is developed for the android mobile platform using Android Studio and Java. The UI is designed to be responsive across various android device screen size and resolutions. User interactions like data submission to Firebase. Error message is displayed for invalid inputs.

3.5.2. System interface requirements

The system integrates with Firebase authentication to manage user login, registrations and session management. The system will facilitate joining classes through a teacher shared link.

3.5.3. Hardware interface requirements

The system run on Android smartphones. The system may require access to standard mobile hardware components such as touch screen for user input and network connectivity for database connectivity.

3.6. Summary

Requirement Analysis provided an insightful overview of project. We analyzed the requirements that are required by our project by using several requirement identifying techniques i.e. use case diagram, story board etc. We also provided a detailed overview of our use cases, which clearly describe each use case along with its activities. After describing the use cases, we also explained the FRs of our system that must be fulfilled to complete system's functionality. By describing these requirements in a structured format, we are confident that no ambiguity is left in the requirement analysis of our project and of course, we cannot forget the NFRs of any project, so we have described them as well, along with the external interface requirements. The project is an Android application developed using java within Android Studio. The Firebase is used as the backend as a service for its real time database and authentication. The non-functional requirements focus on reliability, usability and security. The external interface is the android user interface, the Firebase APIs for data management and standard mobile hardware component.



Chapter 4

Design and Architecture



4. System Design

4.1 Design Considerations

Assumptions

Students, teachers, admins and parents will have a stable internet connection to access the app and manage information. It will provide accurate and up-to-date course schedules, student records and announcements. The application will be used primarily on Android devices and will expand in the future for iOS devices. The payment gateways for fee submission will be functional and available all the time.

Dependencies

Android Studio and other libraries for the development of apps. Third-party payment gateways will be used for transactions (fee submission). The app's backend system uses a database to keep all its information organized (student grades, teacher schedules, etc.). Google Maps to provide location information for the physical campus/branches.

Limitations

The primary limitation is that Kips Coaching Kharian is a Kharian-based service and physical classes are conducted only in Kharian City. The app requires an active internet connection to work properly. Payment failures may arise due to issues with third-party payment gateways.

Risks

There is a risk, if the project does not satisfy the requirements of the management team at Kips Coaching Kharian. Users (students, parents, teachers, admin) can cause errors by entering incorrect data or selecting the wrong options. If a user cancels a subject or requests a refund at a time that is against the terms and conditions, it can cause conflicts. The project might face resistance from traditional administrative staff who do not want to upgrade to the online system.

4.2 Requirement Traceability Matrix:

Traceability Matrix:

Table 59 Traceability matrix

Requirement ID	Requirement Description	Design specification	Module	Code	Test ID
R1	Admin Login	Login component	Admin	AdminLoginActivity.java	TC-01
R2	Reset Password	Reset password	Users	ResetPassword.java	TC-02



		component			
R3	Add, update and delete users	Manage user	Admin	ManageUserActivity.java	TC-03
R4	Approve student and parent registration	Manage student and parent registration	Admin	ApproveRegistrationActivity.java	TC-04
R5	Manage announcements	Announcement manager	Admin	AnnouncementManagerActivity.java	TC-05
R6	View system report	System report viewer	Admin	SystemReportActivity.java	TC-06
R7	Record fee payment	Fee payment manager	Admin	FeePaymentActivity.java	TC-07
R8	Student Register	Register component	Student	StudentRegisterActivity.java	TC-08
R9	Student Login	Login component	Student	StudentLoginActivity.java	TC-09
R10	View enrolled subjects	Enrolled subjects viewer	Students	EnrolledSubjectsActivity.java	TC-10
R11	Download study material	Study material downloader	Student	StudyMaterialDownloader.java	TC-11
R12	View and attempt quizzes and assignments	Quizzes and assignment viewer	Student	QuizAssignmentViewerActivity.java	TC-12
R13	View results and fee history status	Result and fee history viewer	Student	FeeHistoryActivity.java	TC-13
R14	Receive Notification for fee payment	Notification server	Student	FeeNotificationsActivity.java	TC-14
R15	Receive announcements	Notification server	Student	NotificationActivity.java	TC-15
R16	Attempt quizzes	Quizzes component	Student	QuizActivity.java	TC-16
R17	Teacher login	Login component	Teacher	TeacherLoginActivity.java	TC-17
R18	Upload study material	Study material uploader	Teacher	StudyMaterialUploaderActivity.java	TC-18
R19	Create quizzes	Quizzes handler	Teacher	QuizActivity.java	TC-19
R20	Create and share link for	Online classes manager	Teacher	OnlineClassActivity.java	TC-20



	online classes				
R21	Manage student performance	Student performance manager	Teacher	ManageStudentActivity.java	TC-21
R22	View and upload assignment	Assignment viewer	Teacher	AssignmentActivity.java	TC-22
R23	Receive announcement/notifications	Notification server	Teacher	AnnouncementActivity.java	TC-23
R24	Communication with parents	Communication with parent manager	Teacher	CommunicationwithparentsActivity.java	TC-24
R25	Parent registration	Registration component	Parent	ParentregisterActivity.java	TC-25
R26	Parent login	Login component	Parent	ParentloginActivity.java	TC-26
R27	Link to child account	Child account manager	Parent	LinktochildActivity.java	TC-27
R28	View child account	Child account viewer	Parent	Childaccountviewer.java	TC-28
R29	View child result	Result viewer	Parent	Resultviewer.java	TC-29
R30	Fee payment and history	Fee History manager	Parent	Feehistoryviewer.java	TC-30
R31	Fee payment notification	Notification server	Parent	NotificationActivity.java	TC-31
R32	Give review and feedback	Feedback module	Student, parent	FeedbackActivity.java	TC-32
R33	View announcements	Announcement manager	Users	Announcementviewer.java	TC-33
R34	Logout	Logout manager	Users	LogoutActivity.java	TC-34



2. Sequence Diagram

The sequence diagram for KIPS Coaching Kharian describes the flow of interactions between users (like students, parents, teachers, admin) and the KIPS Coaching system components. It covers different scenarios crucial to classes schedule such as searching for subjects, selecting subjects, completing registration through fee payment, manage quizzes and assignments for students, maintaining schedule, generate reports and admin functionalities like manage fee payment process and sends new updates through notifications. The diagram outlines the step-by-step communication between system components (like the student interface, teacher interface, parent interface, database and payment gateway) to execute these specific actions.

Admin Sequence diagram

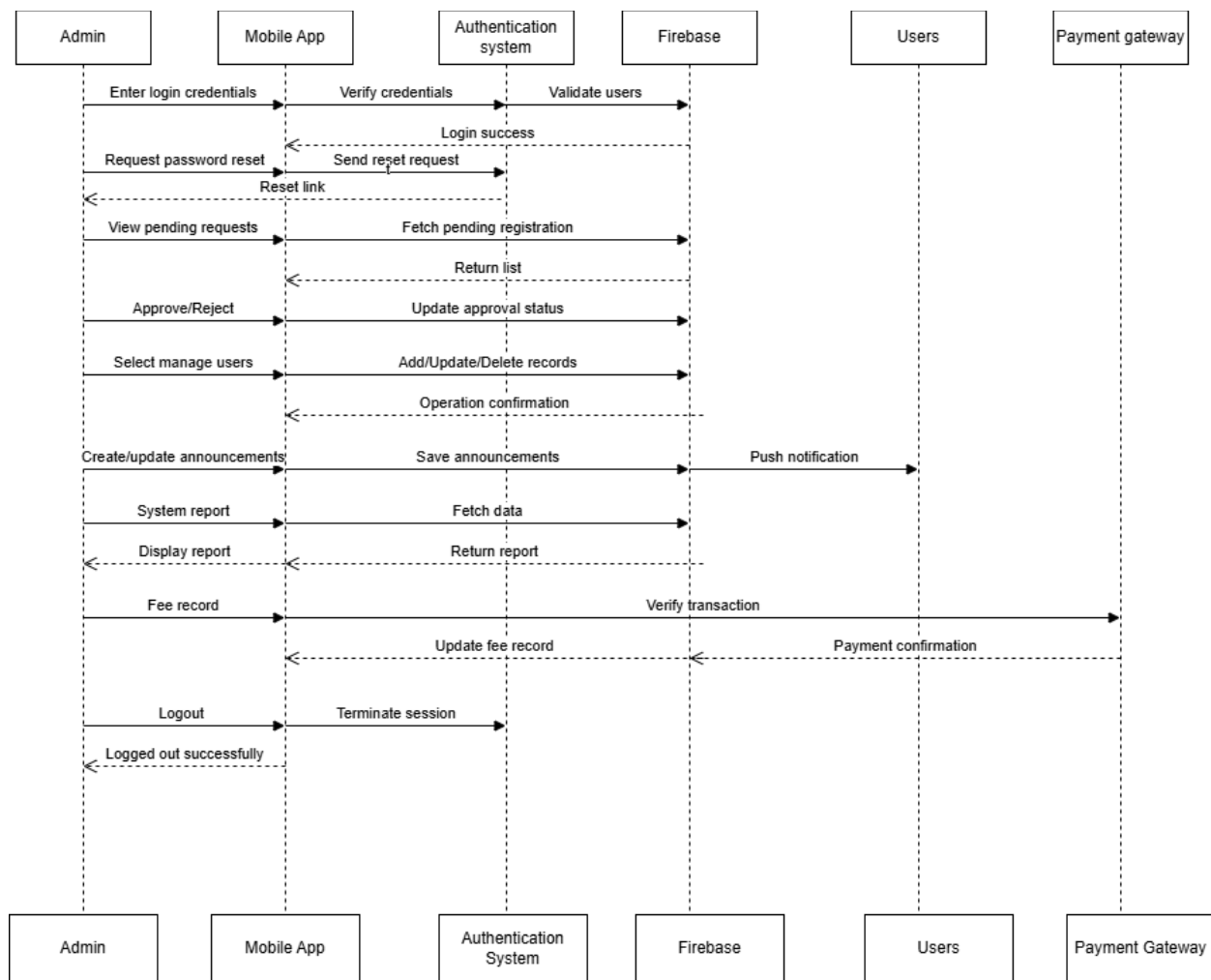


Figure 5: Sequence Diagram (Admin)



Student Sequence diagram

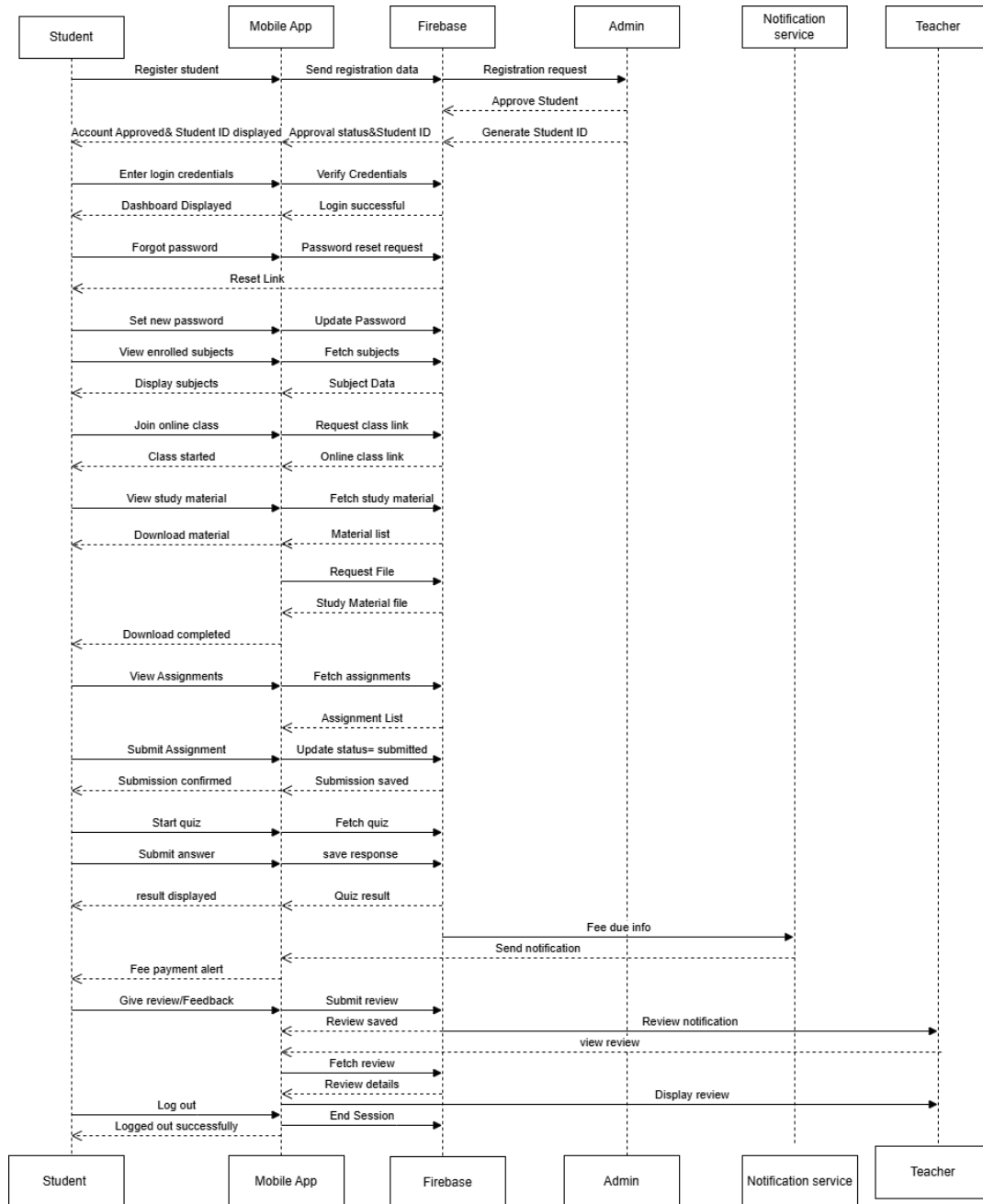


Figure 6: Sequence Diagram (Student)



Teacher Sequence diagram

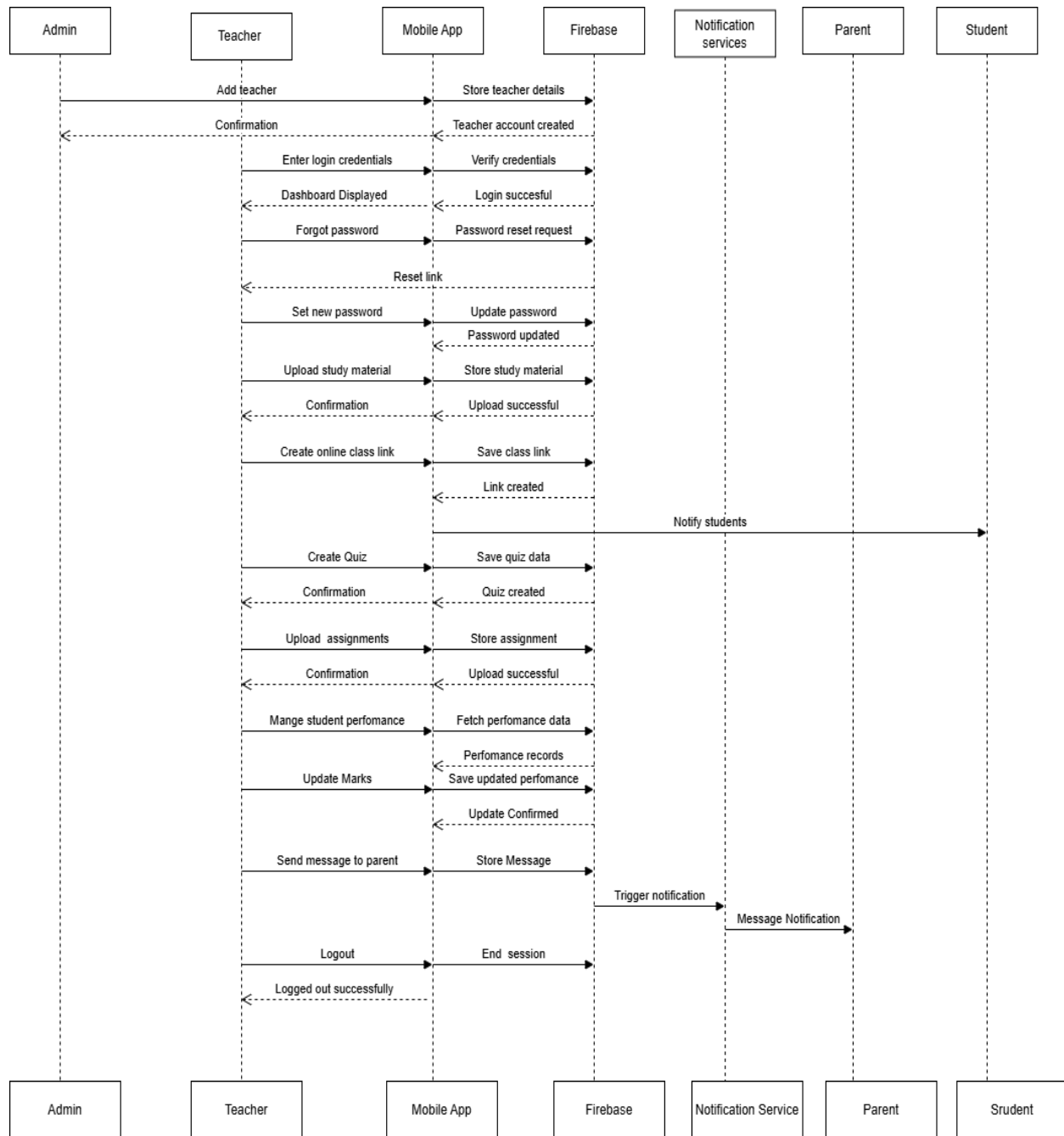


Figure 7: Sequence Diagram (Teacher)



Parent Sequence Diagram



Figure 8: Sequence Diagram (Parents)



3.State Transition Diagram

This diagram shows the different states of the KIPS coaching app and the transitions between them. It covers all the student's login from initially opening the app and to register an account, fees payment, timetable checking, view assignments, quizzes and successfully logout from the app.

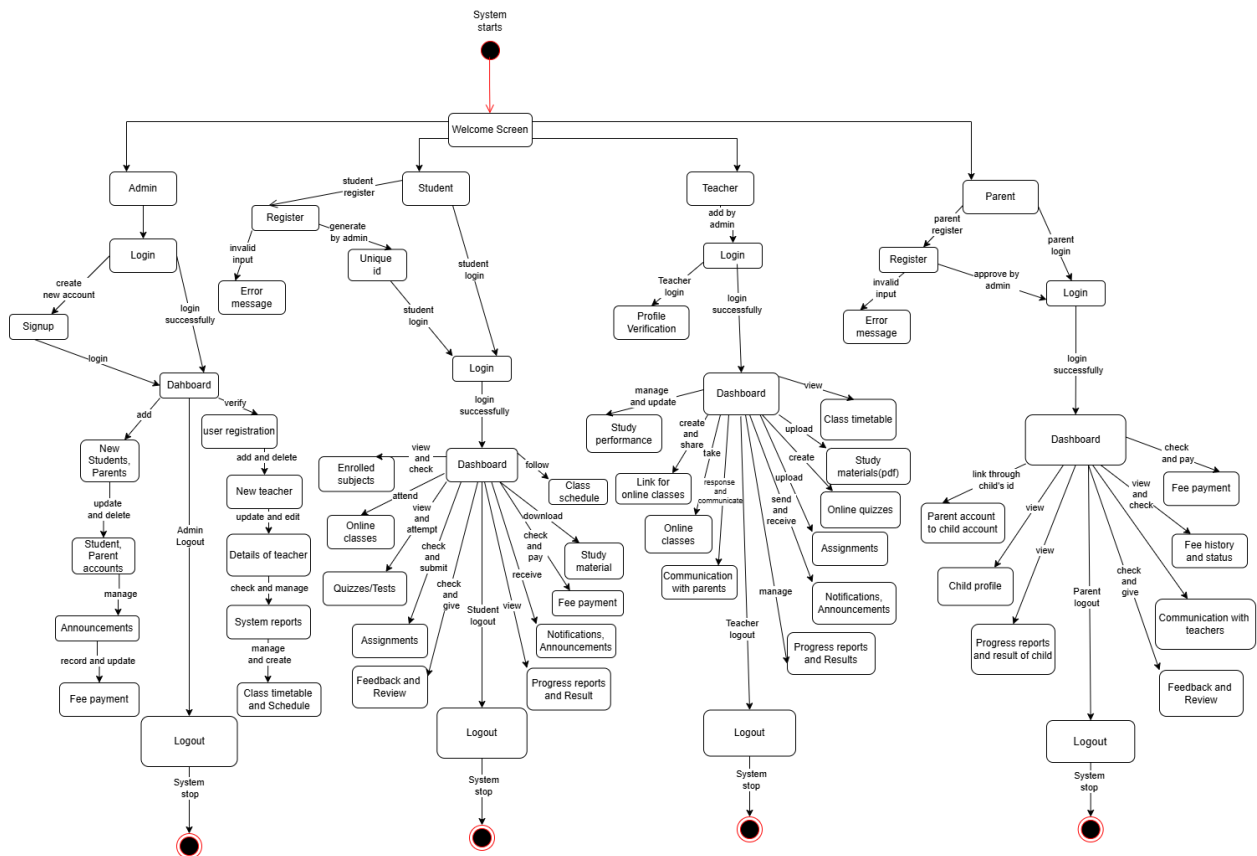


Figure 9: State Diagram

4.4 Architectural Diagram

The KIPS Coaching App is built using a 3-tier structure that includes the User Interface (Frontend), Business Logic (Backend) and Database Layer. The frontend is the mobile app that students interact with.

It allows students to login, register, view schedules, access study materials, view assignments, view quizzes, check results, check fee status and notifications. Enables teachers to manage class schedules, access study materials, upload assignments and quizzes, student progress and send announcements. Provide admins with tools for user management (students, teachers, parents),



course management, fee tracking and announcements. Allows parents to monitor their child progress, grades, fees and communication with teachers.

When a user does something in the app, the backend takes care of the main work, like checking login details, fees handling, assignments and quizzes. This backend then talks to the database, which is built using firebase. The database saves important data. The backend also connects other services like fee payment (such as payfast) to send messages and complete payments safely. This keeps everything clean and organized.

1.Tier Architecture Diagram

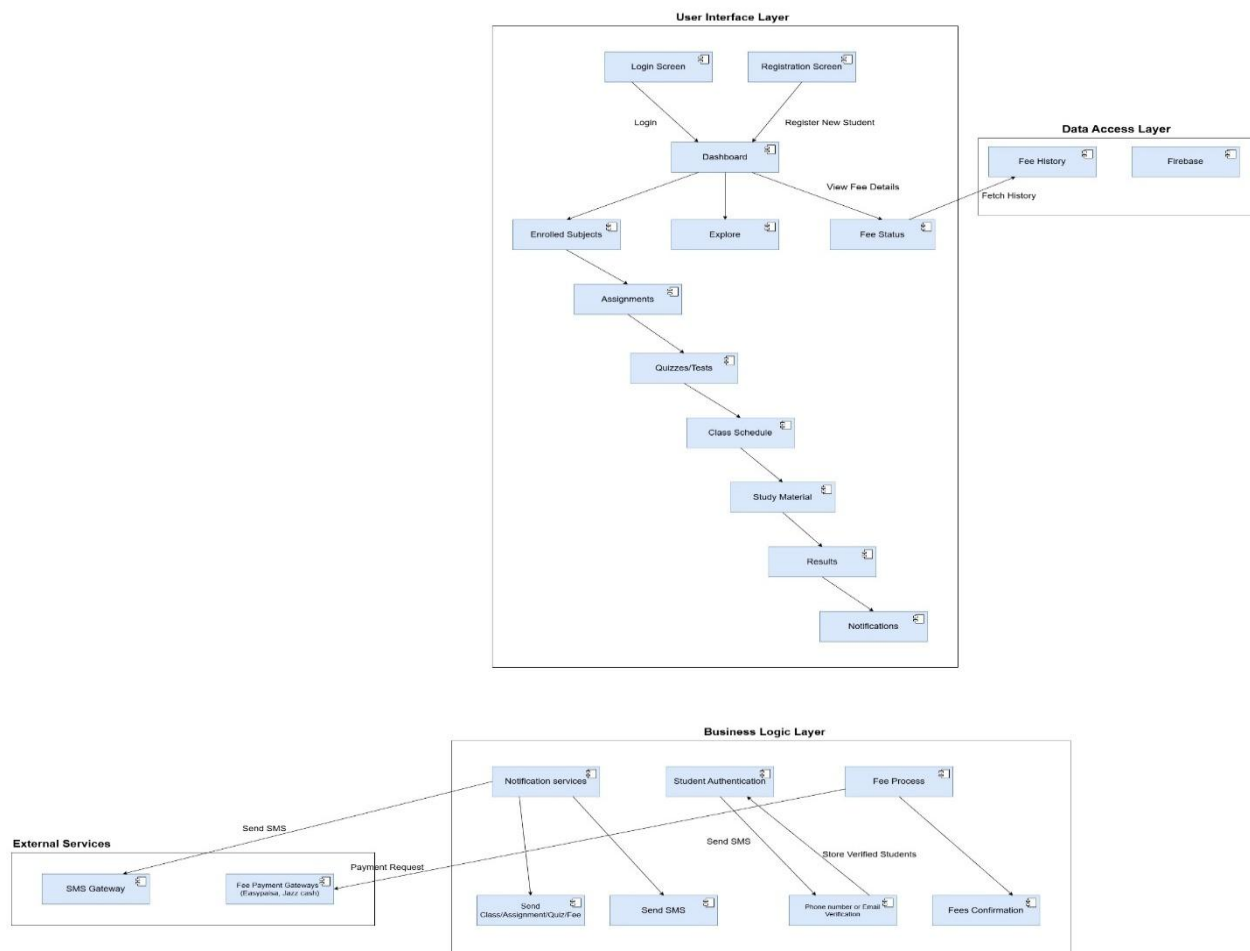


Figure 10: Architecture Diagram



2.UML component diagram:

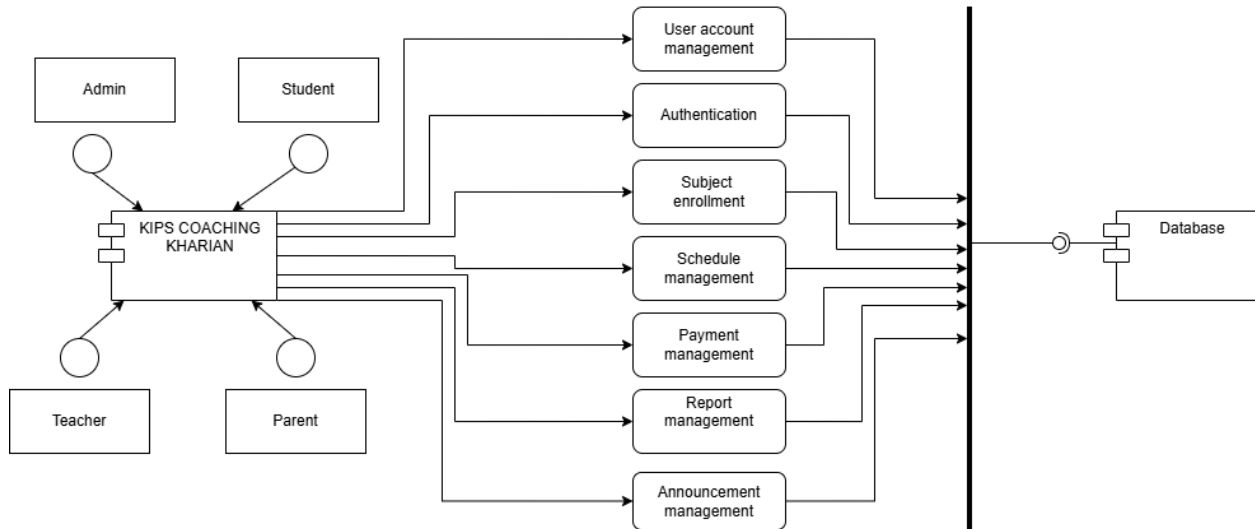


Figure 11: UML component diagram

4.5 Data design

Data design shows how the information in the KIPS COACHING KHARIAN is stored, organized and managed. It uses firebase to store and organize data.



Erd Schema:

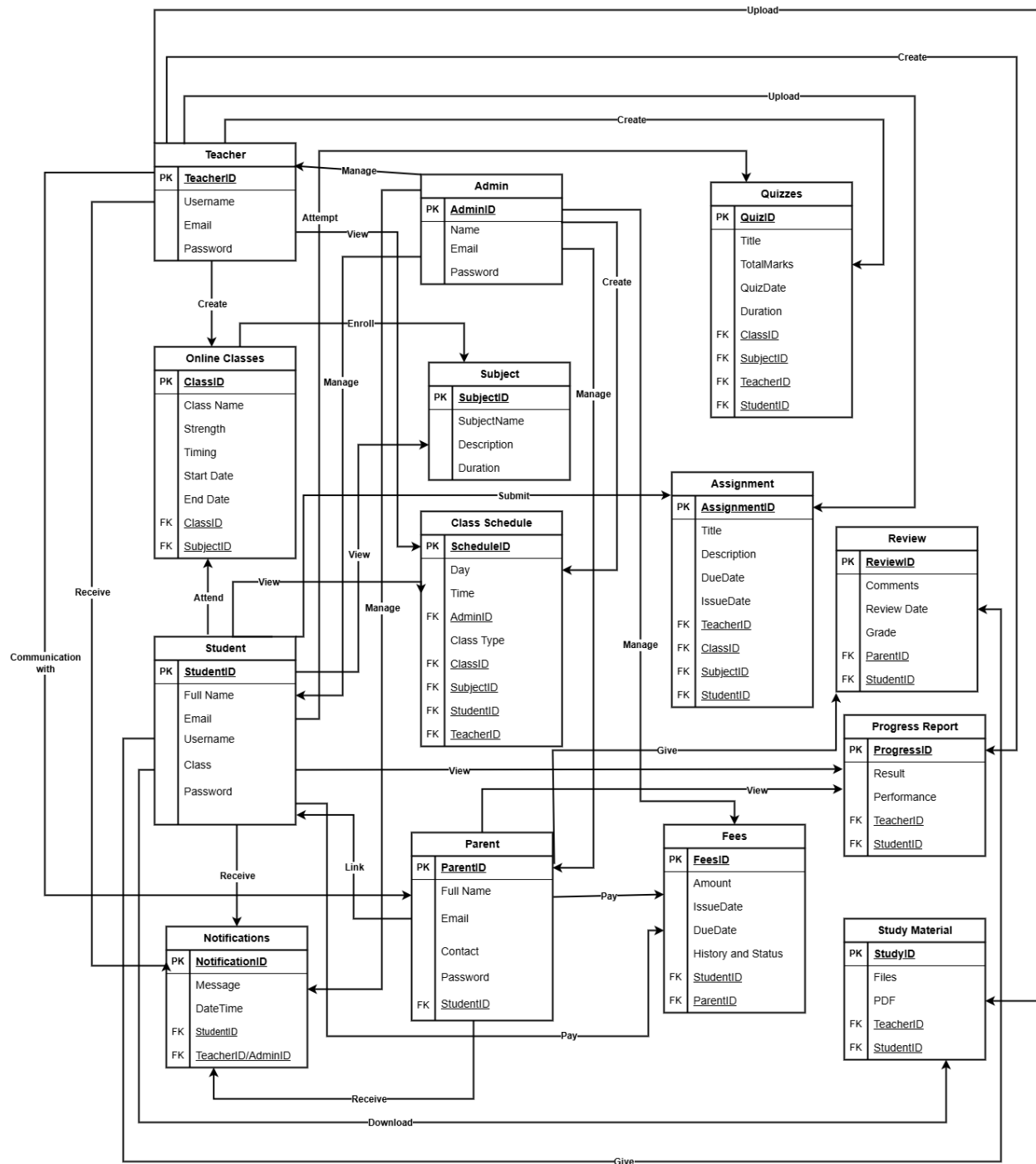


Figure 12: ERD Schema



2.Data dictionary

This table contains the alphabetical list of system entities with their type and descriptions for **KIPS COACHING KHARIAN APP**.

Table 60: Data Dictionary

Terminology	Data type	Description
Amount	Decimal (10,2)	Fee amount
Announcement_description	Text	Announcement details
Announcement_id	Int	Announcement unique ID
Announcement_title	Varchar (150)	Announcement title
Assignment_id	Int	Assignment unique ID
Assignment_marks	Int	Assignment obtained marks
Assignment_status	Varchar (20)	Pending / Submitted
Class	Varchar (50)	Student class
Created_at	Datetime	Account creation date
Due_date	Date	Assignment deadline
Email	Varchar (150)	User login email
Fee_id	Int	Fee record ID
File_url	Varchar (255)	Study material or assignment file
Last_login	Datetime	Last login timestamp
Login_id	Int	Login record ID
Material_id	Int	Study material ID
Message_id	Int	Communication message ID
Name	Varchar (100)	Name of Admin/Teacher/Student/Parent
Notification_id	Int	Notification ID
Notification_message	Text	Notification content
Password	Varchar (255)	Encrypted password
Payment_date	Date	Fee payment date
Payment_method	Varchar (50)	Cash / Online
Phone	Varchar (15)	Contact number
Quiz_id	Int	Quiz ID
Quiz_marks	Int	Obtained quiz marks



Remarks	Varchar (255)	Teacher remarks
Register_id	Int	Registration ID
Role	Varchar (50)	Admin / Teacher / Student / Parent role
Status	Varchar (20)	Paid / Unpaid
Student_id	Int	Student unique ID
Subject	Varchar (100)	Teaching subject
Teacher_id	Int	Teacher ID
Title	Varchar (150)	Quiz / Assignment / Material title
Total_marks	Int	Quiz total marks
Upload_date	Datetime	Material upload date
User_id	Int	General user ID
User_name	Varchar (100)	Registered user name
User_type	Varchar (20)	Admin / Teacher / Student / Parent

4.6 User interfaces design

4.6.1 Screen Images

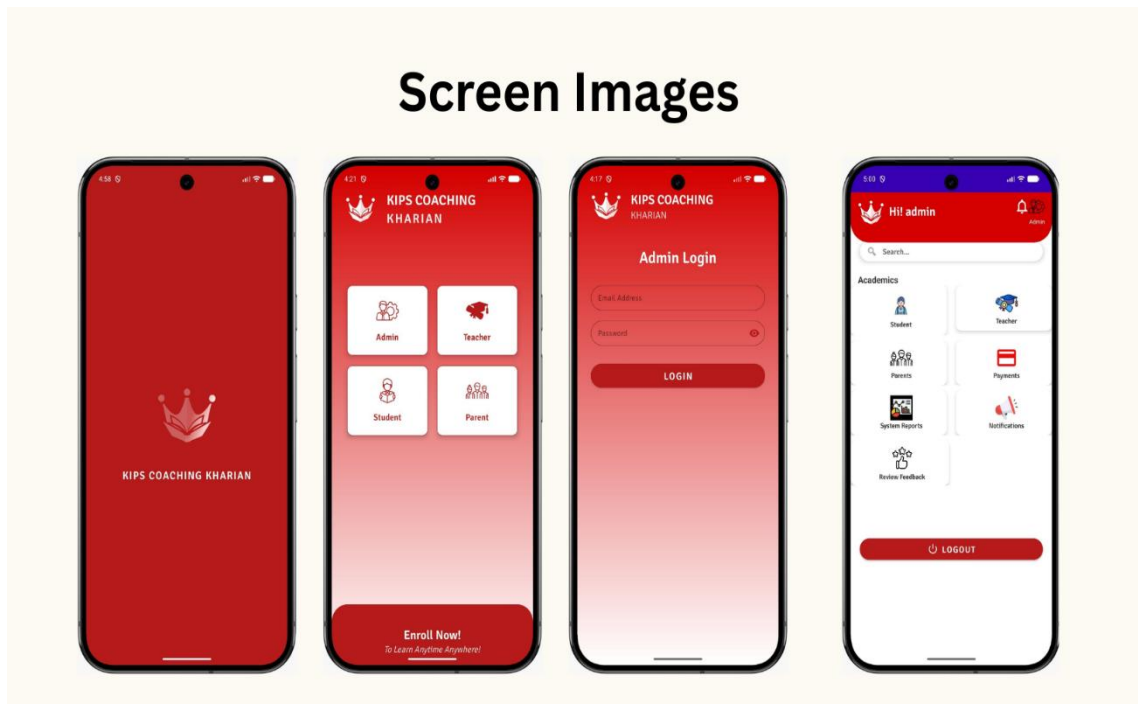


Figure 13: Screen image



Screen Images

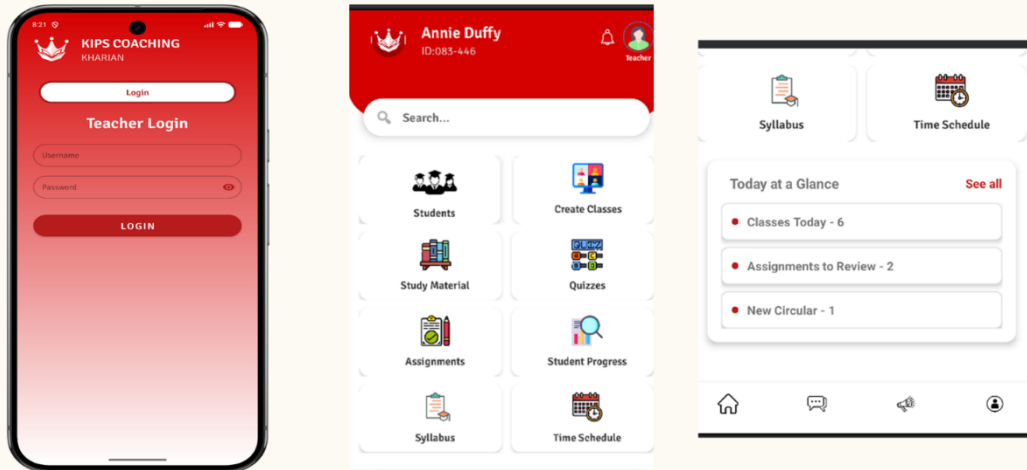


Figure 14: Screen image

Screen Images

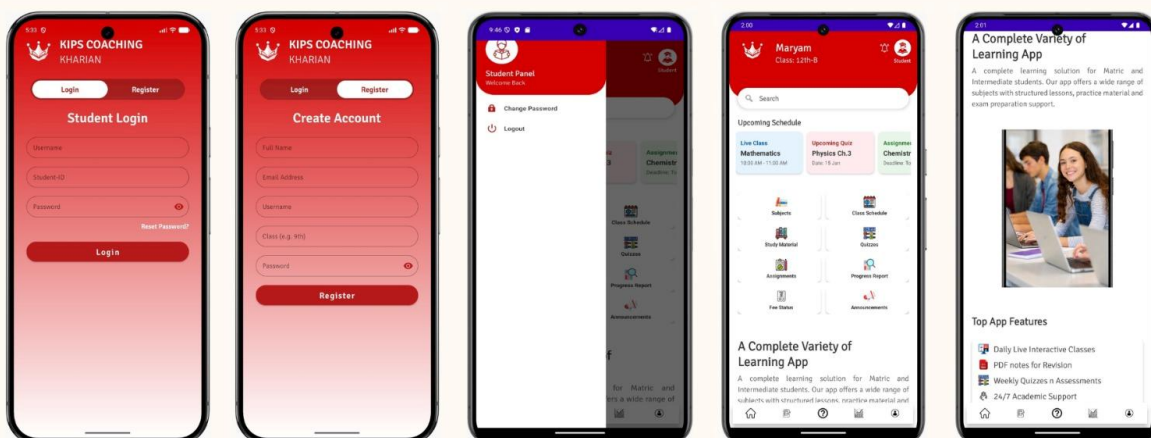


Figure 15: Screen image



Screen Images

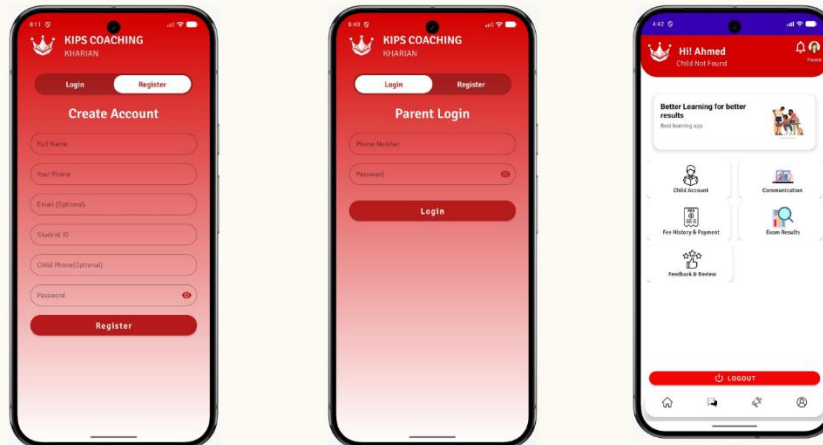


Figure 16: Screen image

4.6.2 Screen Objects and Actions:

This section details the screen components and the functions they perform for use cases:

Table 61: Screen object (Live classes)

Screen Object	Action
Teacher dashboard	Create account and access teacher dashboard then manage live classes section
Create live class	Teacher creates live class
Start/end live class	Teacher starts the live class and end it after completion
Student dashboard	Student logs in and view scheduled live class
Attend live class	Student attend live class and exits when the class end



Table 62: Screen object (Notifications and announcements)

Screen Object	Action
Admin dashboard	Access and manage notifications and announcement
Create notification/announcement	Admin create notifications and announcements
Send notifications	Send notifications to student, teacher and parent
View/receive notifications	View/receive notifications and announcements

Table 63: Screen object (Fee payment and confirmation)

Screen Object	Action
Student/parent dashboard	Access the fee payment section
Fee details	Students and Parents view fee amount, due date and payment status
Enter payment details	Provide necessary information
Pay button	Complete the fee payment transaction
Payment confirmation alert	In app confirmation payment notification sent

4.7 Behavioral Model

1.Sequence Diagram

The sequence diagram helped visualize how different parts of the system, like the app, database and payment gateways are work together for booking of guider and hotels.



Admin Sequence diagram

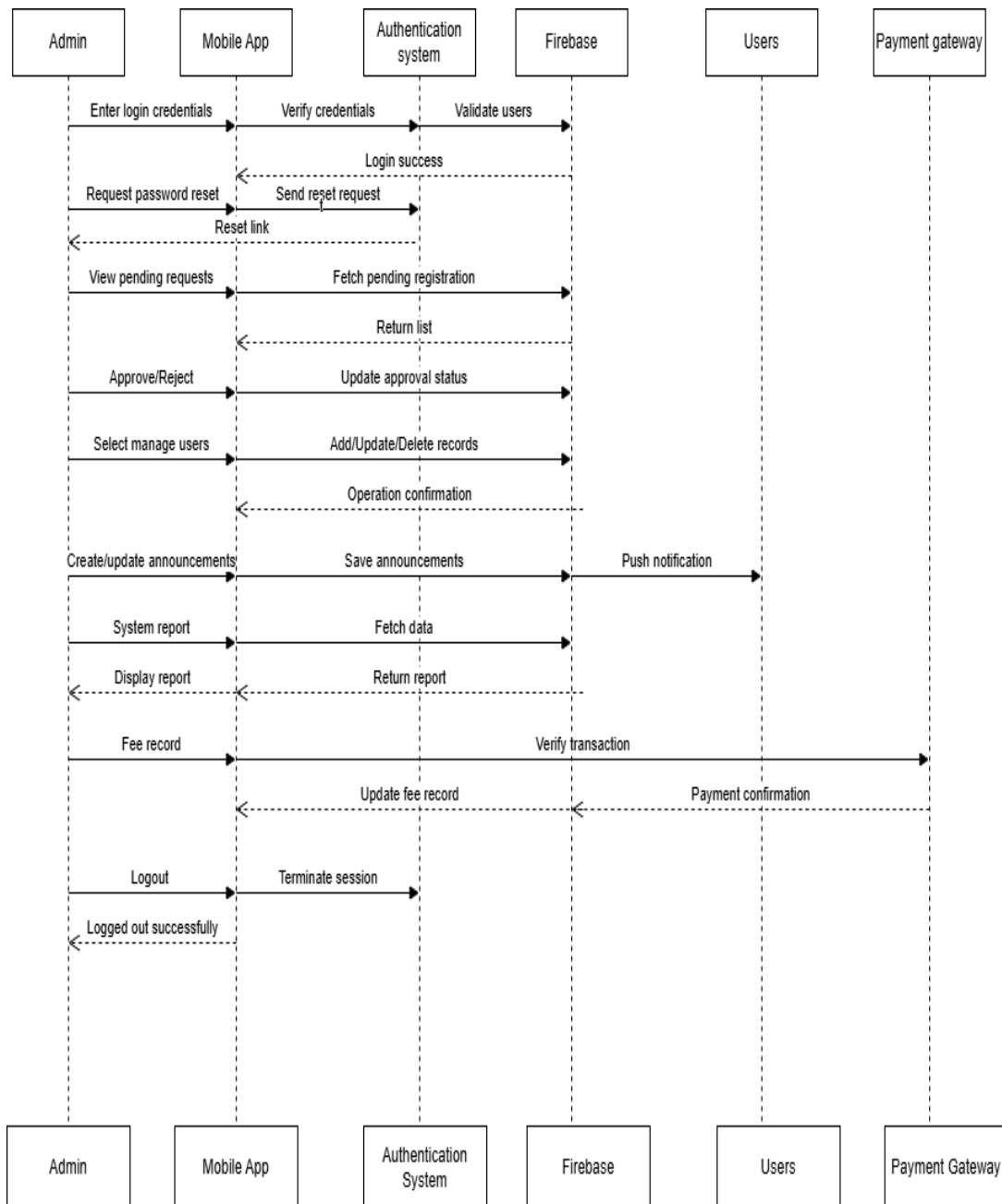


Figure 17: Sequence Diagram (Admin)



Student Sequence diagram

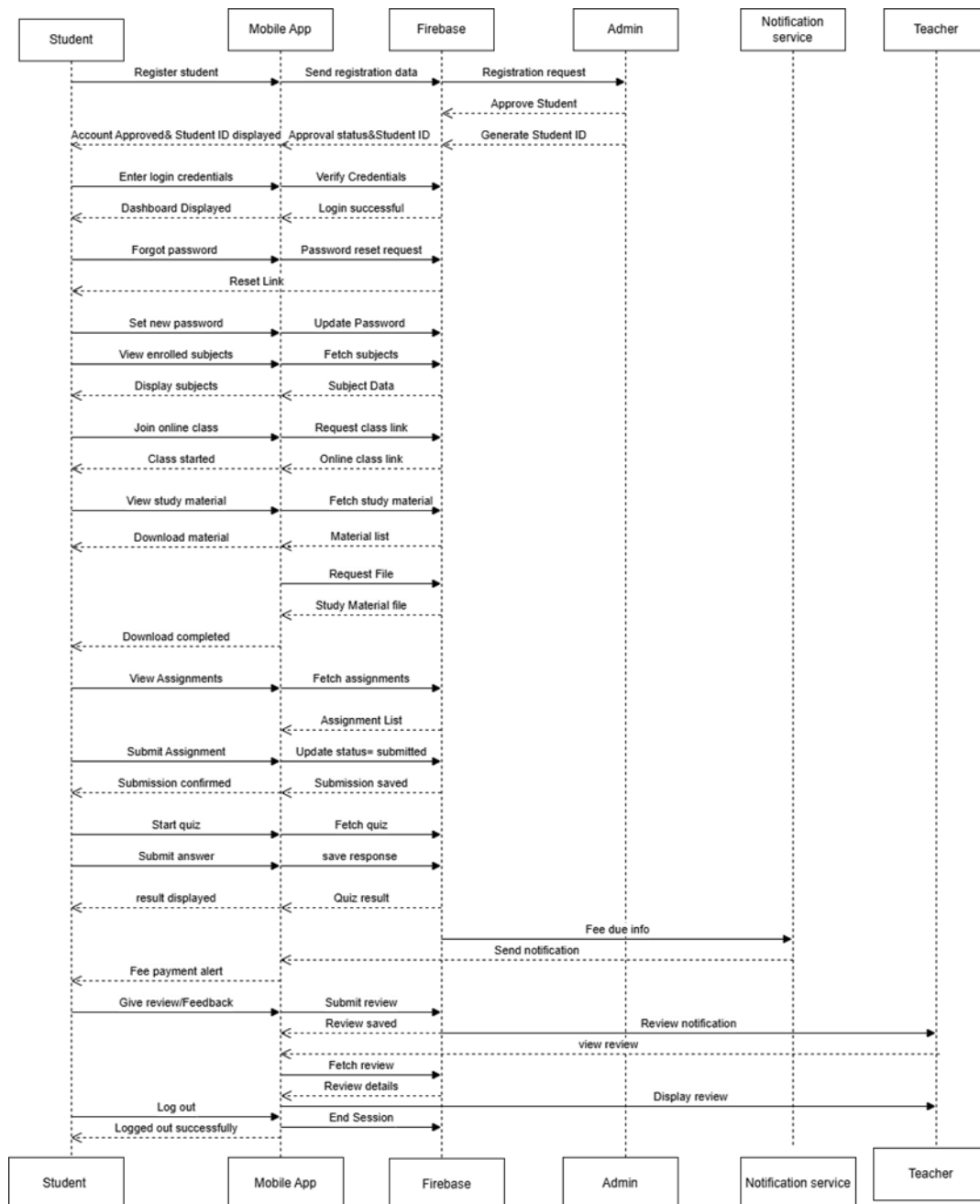


Figure 18: Sequence Diagram (Student)



Teacher Sequence diagram

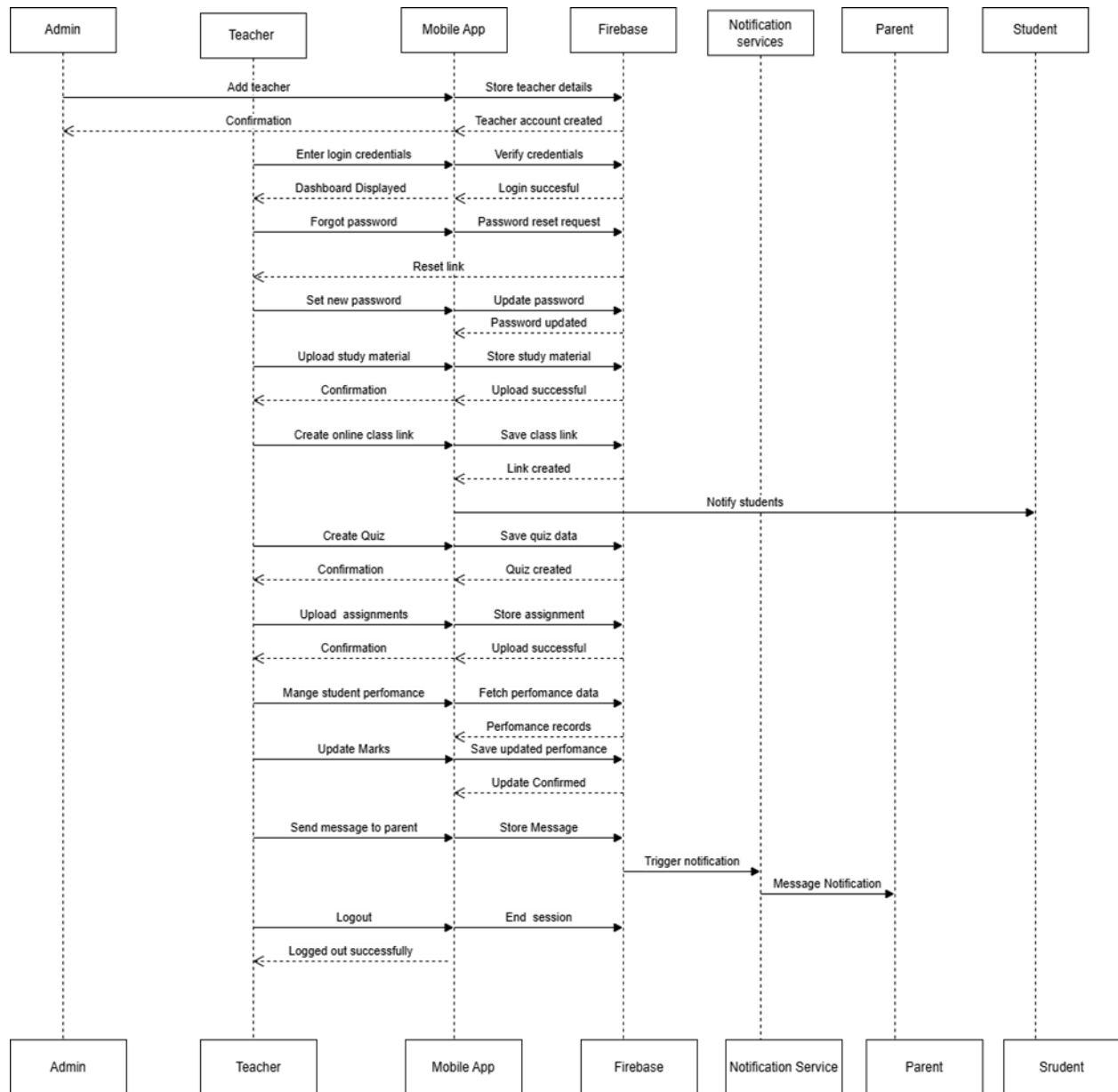


Figure 19: Sequence Diagram (Teacher)



Parent Sequence diagram



Figure 20: Sequence Diagram (Parents)



2.State Transition Diagram

Shows state transitions within the software, especially useful for systems with complex workflows or lifecycle states.

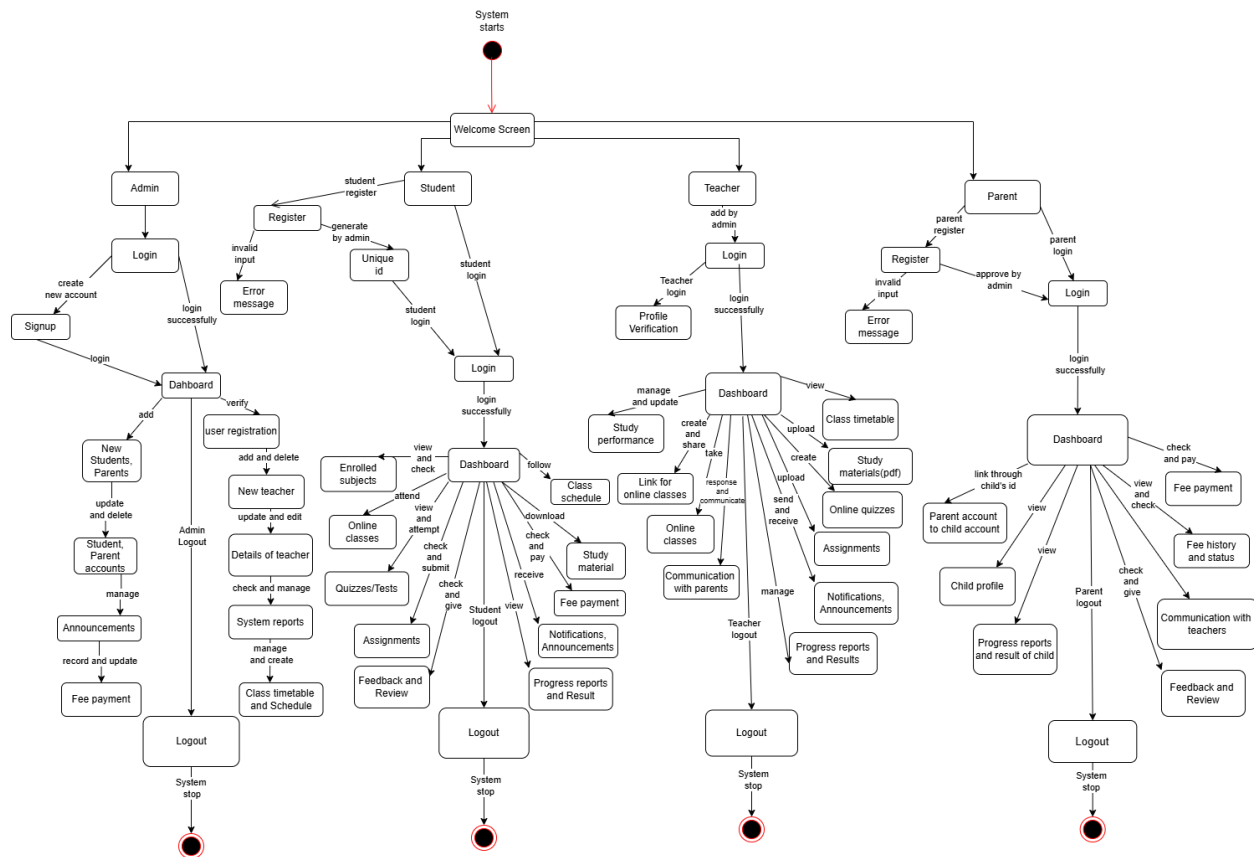


Figure 21: State Diagram

4.8 Design Decisions

We made key design decisions to ensure our system easy to use:

1. Object Oriented Design (OOD)

We use Object- Oriented Programming (OOP) to organize the system in modular and maintainable way. This makes the system easy to upgrade, scale and manage. The KIPS COACHING management system is build using an object – oriented approach, where each object represents real –world entities within the coaching environment, such as Students, Teachers, Subjects,



Assignments and notifications. Each object contains relevant attributes and methods, making the system organized and efficient for future enhancements.

Table 64: Object and attributes

Object Name	Description	Attributes
Admin	Manage overall system including students, teachers, parents, subjects and fees	name, email, password
Student	Represent a students enrolled in KIPS Coaching Kharian	student_id, name, class, parent_contact
Teacher	Represents a teacher or faculty member	teacher_id, name, password, email
Parent	Represent a parents	Parent_id, name, password, email
Subjects	Represents subjects offered by the coaching center	subject_id, name, teacher_assigned_id
Live class	Represents live classes conducted by the teachers	class_id, subject_id, teacher_id, date, time, meeting link, platform, duration
Quiz	Represents quizzes created by the teacher	quiz_id, subject_id, teacher_id, duration, total marks, topics
Assignment	Represents assignments for students	assignment_id, subject_id, teacher_id, student_id, due_date, submission_date, marks_obtained
Notification	System generated notifications for the students, teacher and parent	notification_id, recipient_type, recipient_id, message
Fee	Track student fee payment	fee_id, student_id, amount, due_date, payment_date, payment_method, status
Timetable	Represents daily or weekly class schedules	timetable_id, class_id, day, start_time, end_time



2.Model-View-Controller (MVC) Architecture

We followed the MVC pattern to separate data, business logic and the user interface. This approach keeps the system organized, easy to maintain and user friendly for students, teachers and admins.

3.Database Design and Normalization

We use firebase as the backend database to ensure fast, reliable data storage. The data is normalized to avoid duplication, ensure quick searches and maintain consistency for student records, study materials and fee transactions.

Conclusion

By using OOP, MVC and an optimized database, our KIPS Coaching Kharian provides a smooth and user friendly interface.

Students can easily view subjects, attend online classes, view assignments and fee payments. Teacher can efficiently manage study materials, assignments, quizzes and student performance. Over all the system ensures efficiency, reliability and ease of use for everyone involved.

4.9Summary

This chapter explain how the **KIPS COACHING KHARIAN App** provides a smooth and automated academic management system. The system follows an object- oriented approach, making it well- structures and easy to manage. Using an object oriented approach, the app efficiently manages student records, class schedules, assignments, notifications and fee details. The sequence and state diagram showed how the student, teacher, parent and admin modules interact and how task moves through different stages. Key improvements such as fee alerts, parent monitoring and teacher tools for the uploading quizzes and assignments make the app easy to use, efficient and reliable.

4.10 References

- KIPS Education System

KIPS Virtual Learning Platform.

<https://kips.edu.pk>

Related system reference (Student enrolments and recorded lectures).

- The Educators School System

Online Academic Management System.

<https://educators.edu.pk>

Example of integrated digital education platform.

- Roots International Schools

Roots Learning Management System.

<https://rootsinternational.edu.pk>

Reference for paid and large-scale LMS platforms.